Activity Diagram Documentation

Introduction

This document presents the complete activity diagram set for the Tour and Travel Agency Operations System, organized in two tiers.

Tier 1 - Subsystem overviews: Eight diagrams covering one major subsystem each. These show how activities chain together at a high level and which actors are involved.

Tier 2 - Individual use case diagrams: Thirty diagrams, one per use case. These model every decision branch, actor handoff, alternate path, error condition, and system response for that interaction.

Color coding: blue = customer-initiated actions, purple = administrative/HR actions, teal = system outputs and confirmations, coral = errors and rejections, amber = financial operations, pink = customer experience features, green = reporting. Gray indicates neutral system steps.

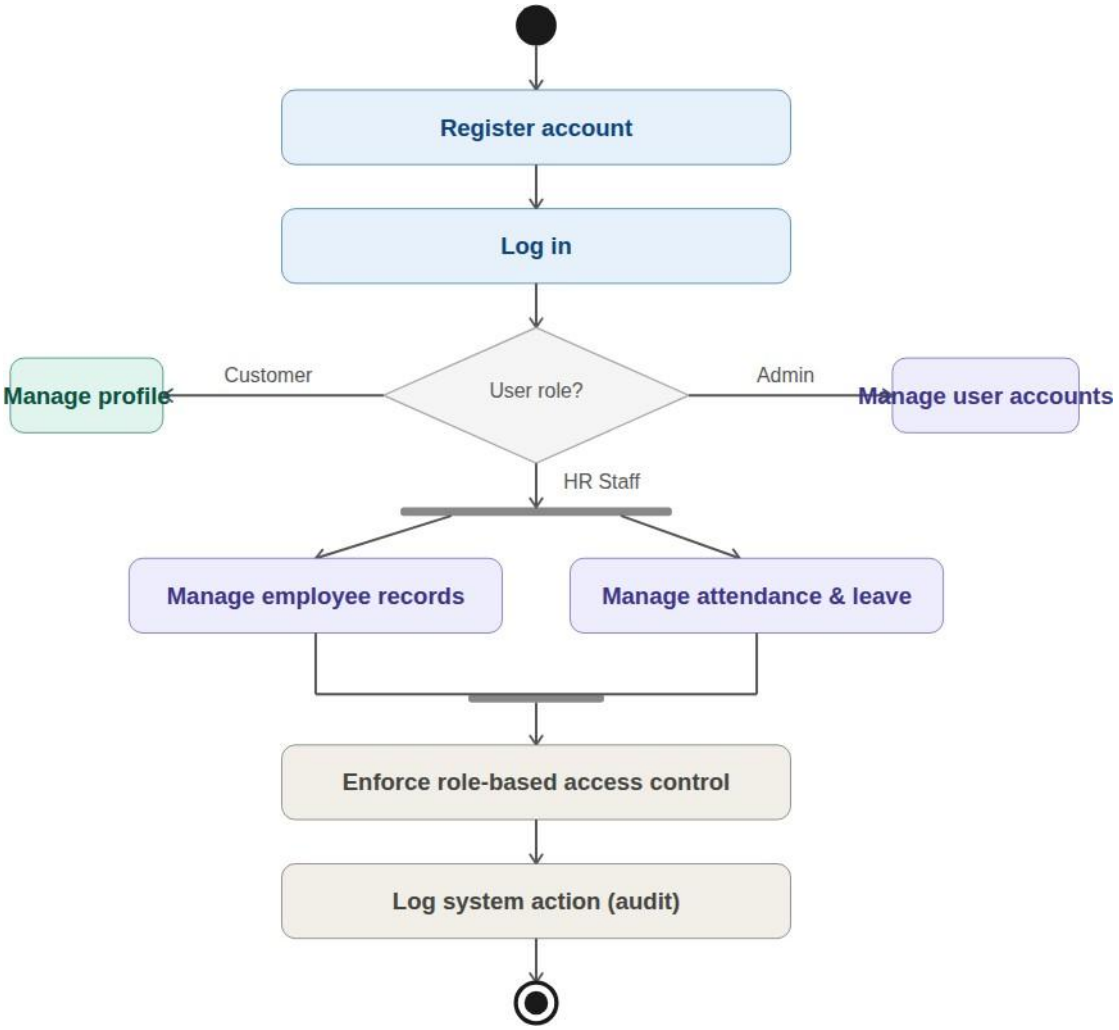
Part 1 - Subsystem Overview Diagrams

Eight diagrams, one per subsystem. They provide breadth of coverage without drilling into every branch of each individual use case.

#	Subsystem	Actors
1	User & Access Management	Customer, Administrator, HR Staff
2	Booking & Reservation Management	Customer, Travel Agent
3	Travel Package Management	Staff, Customer, Travel Agent
4	Cancellation Handling	Customer, Travel Agent, Finance Staff
5	Financial Management	Finance Staff, Manager, Customer
6	Scheduling & Operations	Staff
7	Customer Experience	Customer
8	Reporting & Analytics	Manager, Finance Staff

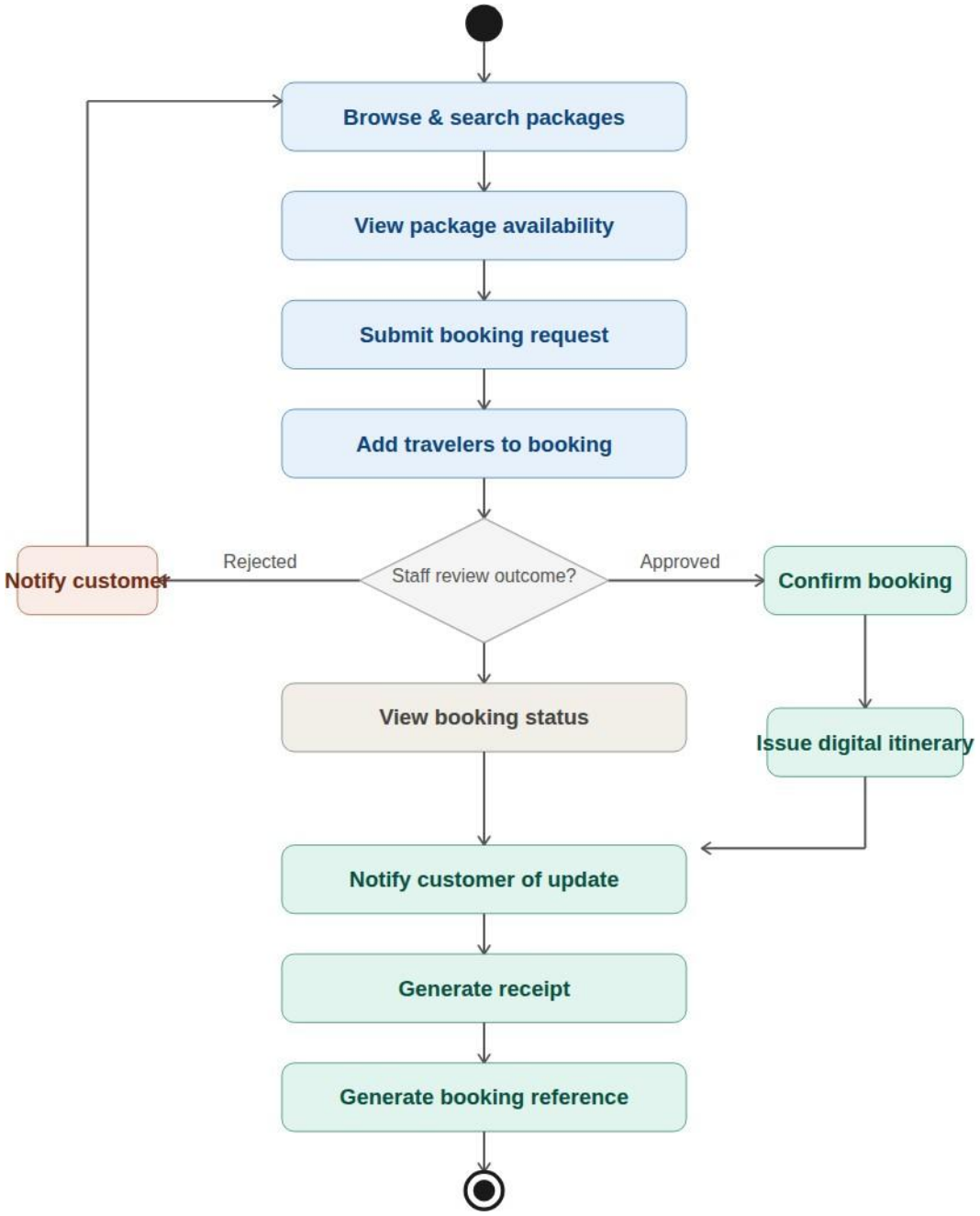
1.1 User & Access Management

Covers account registration, login with role-based routing, customer profile management, administrator account control, and HR management of employee records, attendance, and leave. All events are logged to the system audit trail.



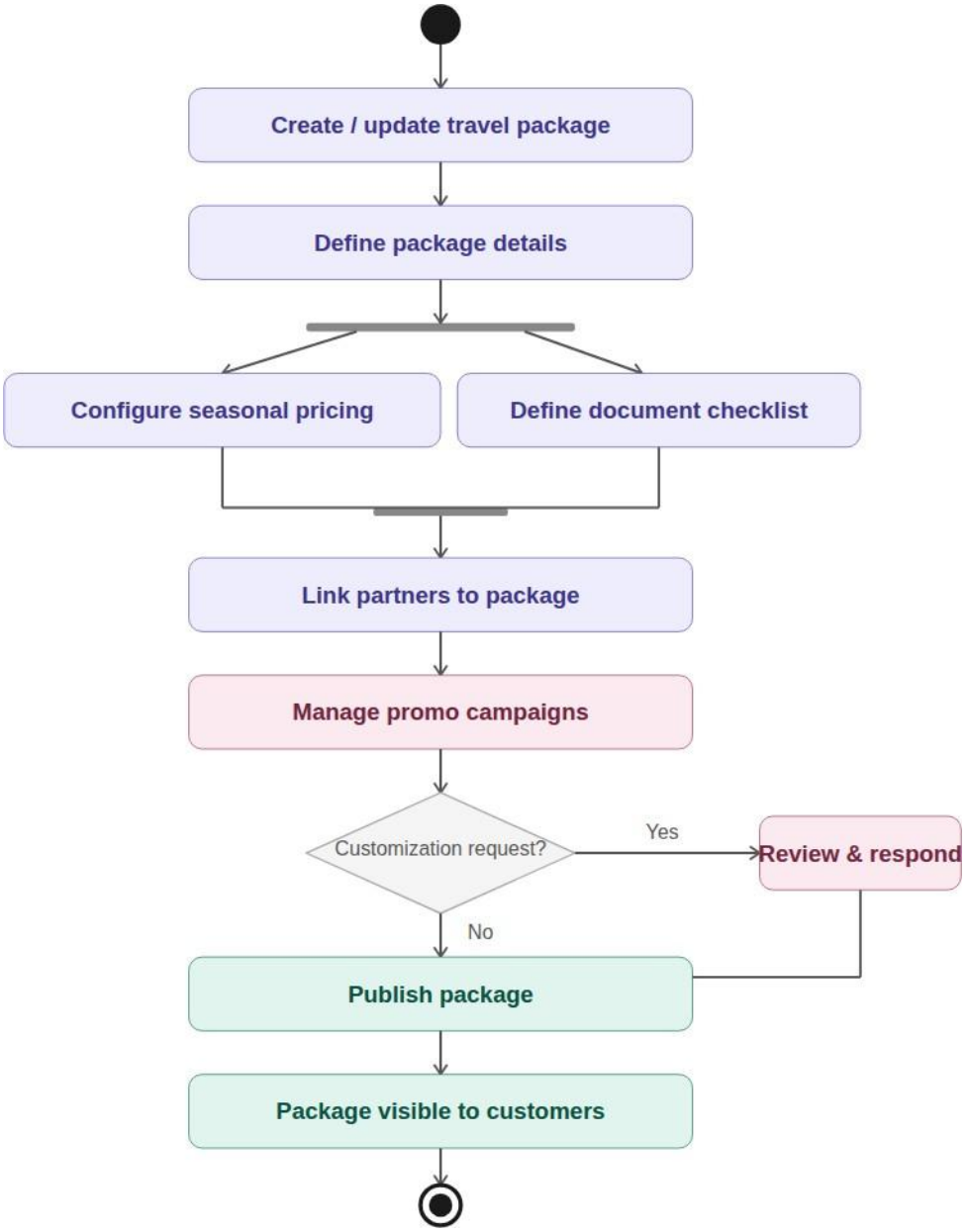
1.2 Booking & Reservation Management

Covers the full booking lifecycle: browsing, availability check, booking submission, traveler registration, staff review (approved / rejected / on hold), itinerary issuance, receipt and reference generation, and customer notification.



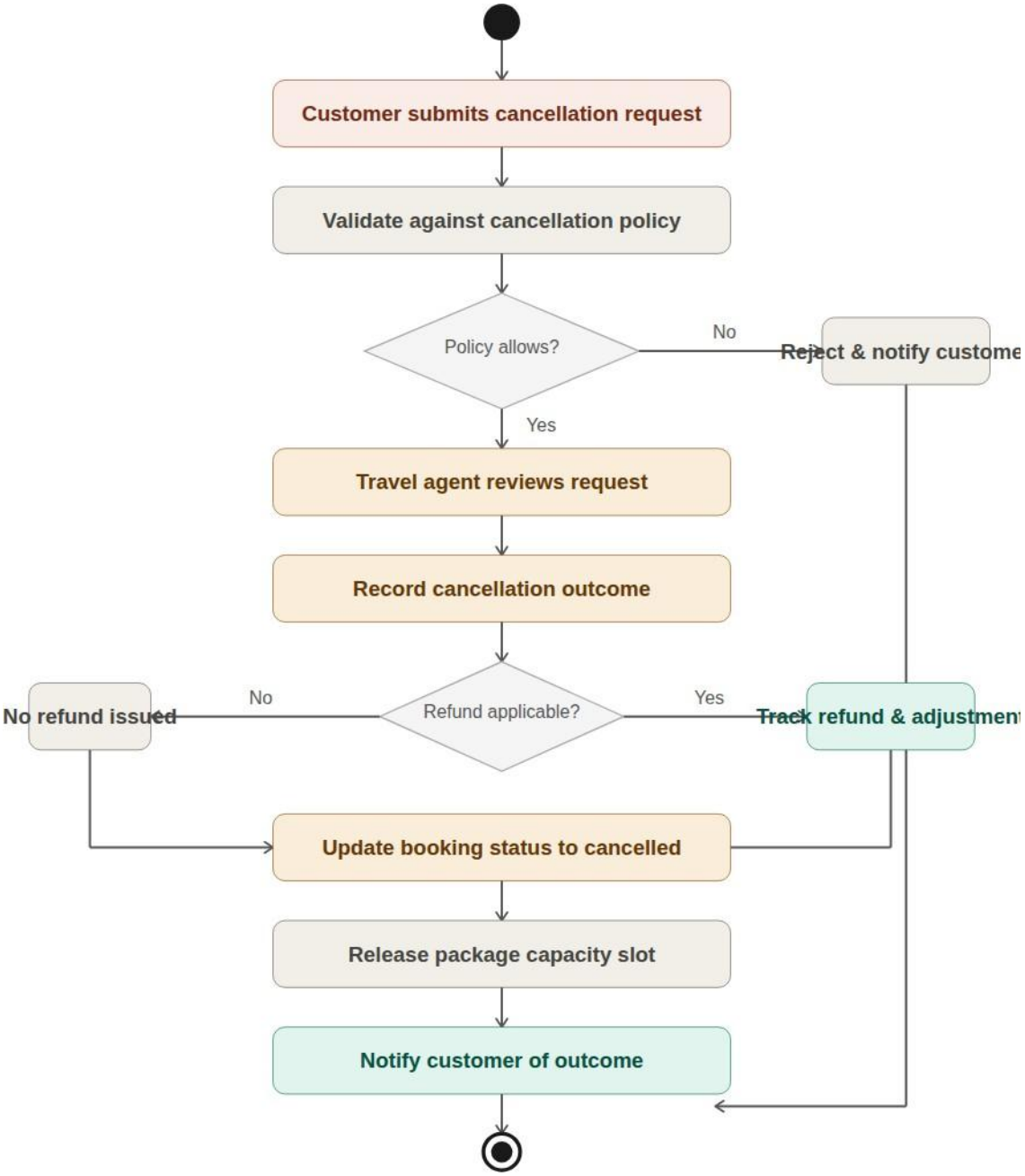
1.3 Travel Package Management

Covers package creation, detail definition, parallel seasonal pricing and document checklist configuration, partner linking, promotional campaign management, and customization request handling before publication.



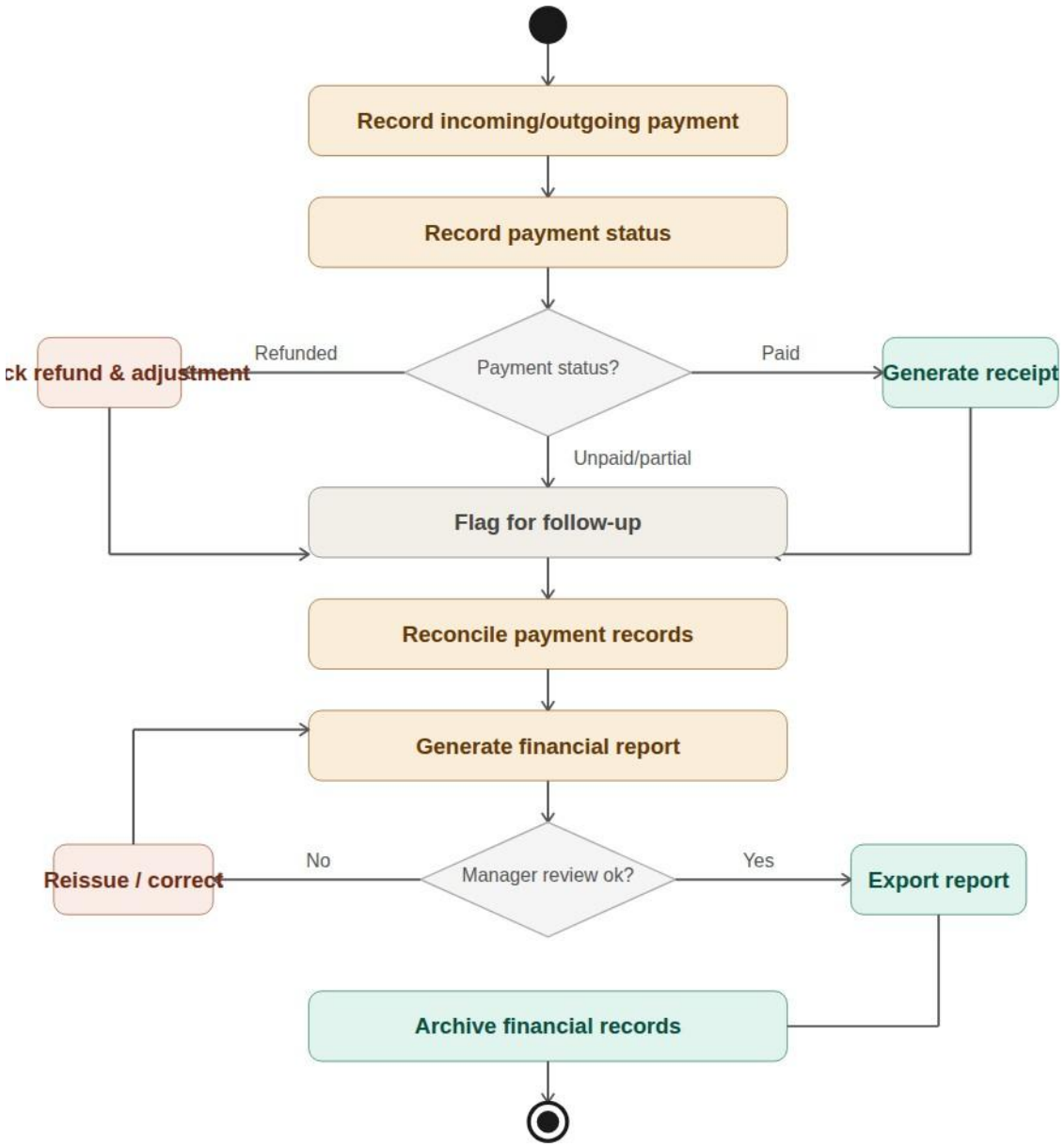
1.4 Cancellation Handling

Covers cancellation submission, policy validation, travel agent review, outcome recording, refund applicability decision, booking status update, capacity slot release, and customer notification.



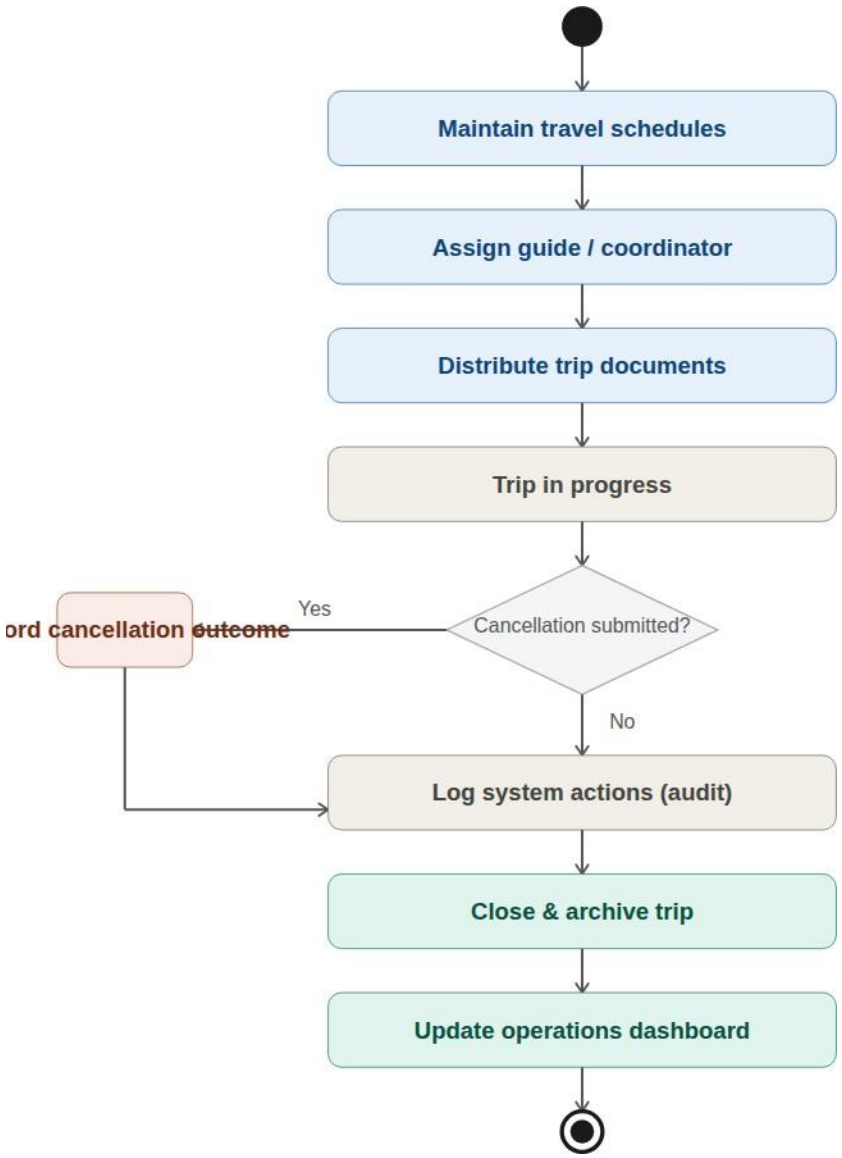
1.5 Financial Management

Covers payment recording, five-way status branching, receipt generation, reconciliation, report generation with manager sign-off, report export, and financial record archival.



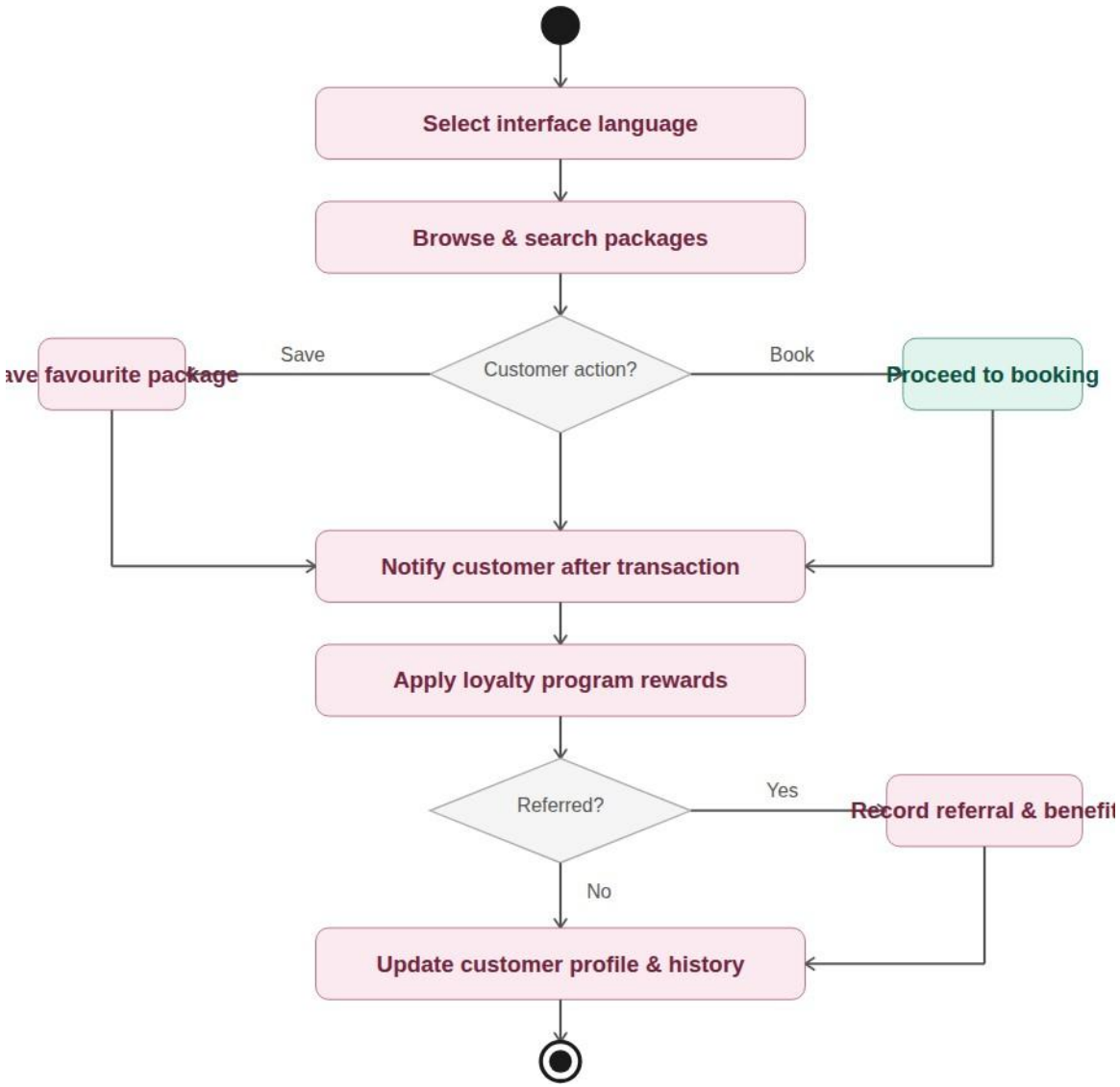
1.6 Scheduling & Operations

Covers schedule maintenance, guide assignment, document distribution, trip monitoring with a cancellation branch, system action logging, trip closure, and dashboard update.



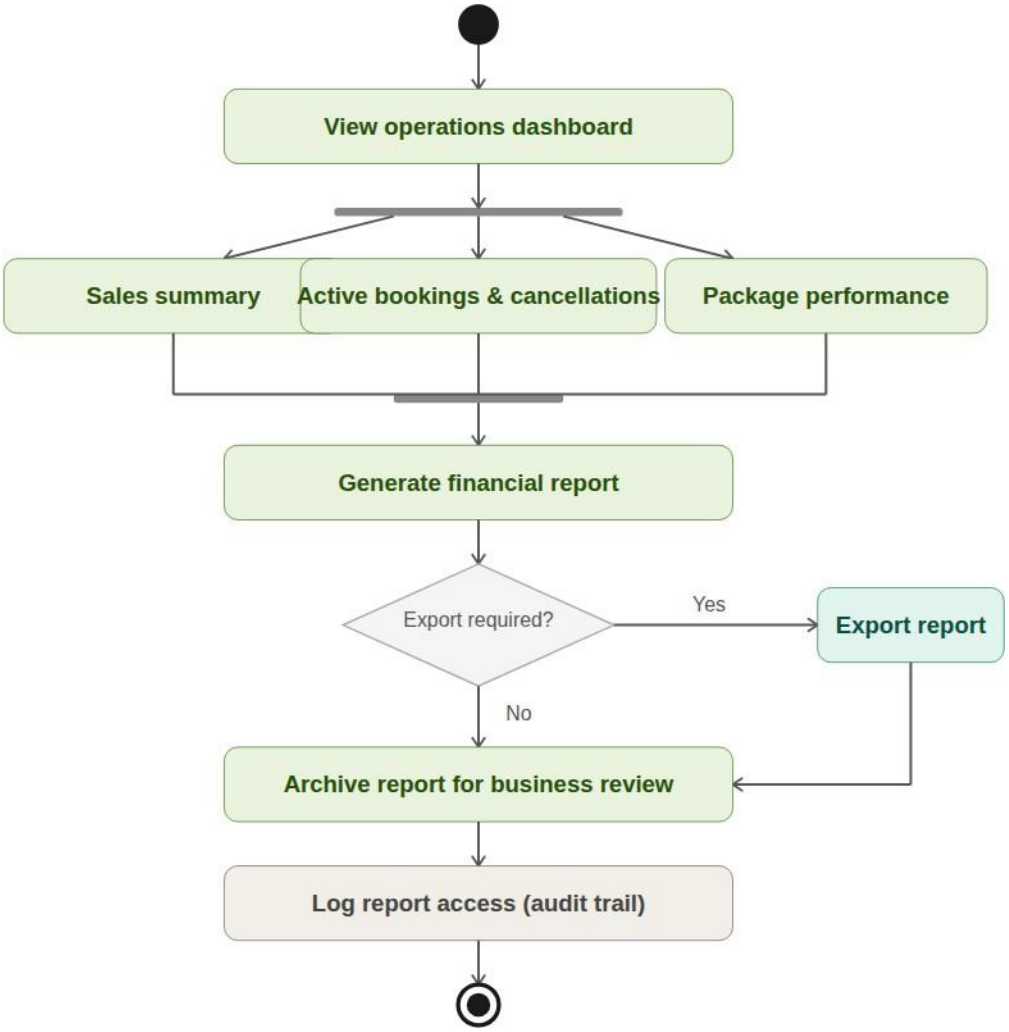
1.7 Customer Experience

Covers language selection, package browsing, saving favourites, proceeding to booking, post-transaction notification, loyalty reward application, and referral tracking.



1.8 Reporting & Analytics

Covers dashboard viewing with parallel data aggregation across sales, bookings, and package performance, financial report generation, optional export, archival, and audit logging of report access.



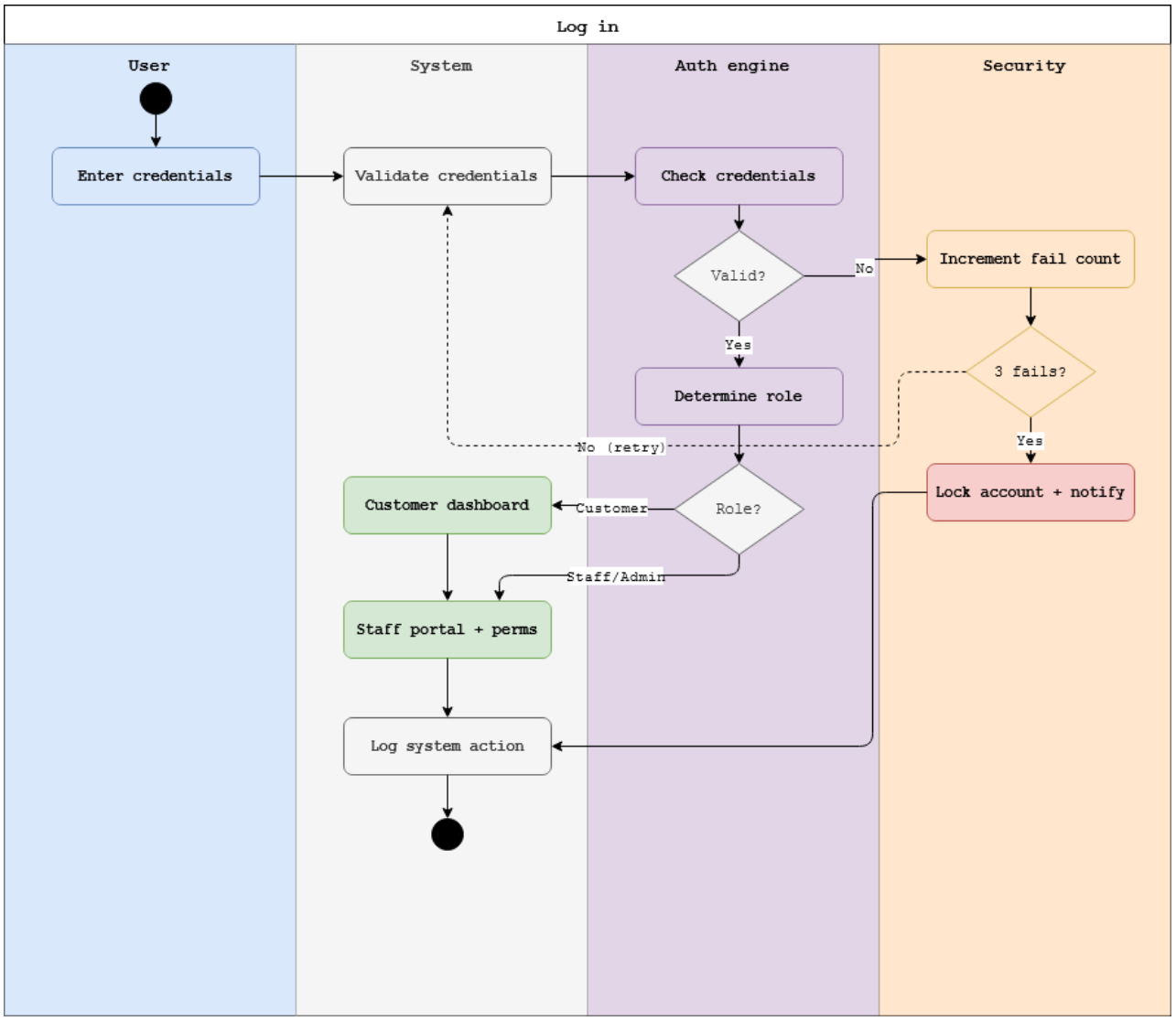
Part 2 - Individual Use Case Activity Diagrams

Use case	What the diagram covers
Log in	Credential validation, lockout, role routing
Register account	Field validation, uniqueness check, account creation
Manage customer profile	Edit info, add traveler, view booking history
Manage user accounts	Create, edit, disable, reactivate
Manage employee records & attendance	Records update, leave request approval
Manage travel packages	Create, update, archive, publish
Define package details	Required field validation, catalog save
Configure seasonal pricing	Date range and discount rule
Define document checklist	Required document linking to package
Manage promo campaigns	Code creation, discount application
Submit customization request	Feasibility review, accept or reject
Submit booking request	Eligibility check, 3-way review decision
View package availability	Capacity display, full vs available state
Issue digital itinerary	Generation and delivery to customer
View booking status	Self-service status display
Add notes / view audit trail	Staff annotation and history log
Submit cancellation request	Policy check, refund decision, notification
Record payment status	Five-status branching, booking update
Generate receipt	Auto and manual issuance
Reconcile payment records	Discrepancy detection and correction
Track refunds & adjustments	Amount recording, financial record update
Maintain travel schedules	Add, edit, remove departure dates
Assign guide / coordinator	Conflict check, confirmation, notification
Close & archive trip	Completion, document distribution, archival
Manage business partners	Add / update partner directory
Link partners to packages	Partner-package association
Browse & search packages	Filter, display results, empty state
Manage loyalty & referral programs	Reward application and referral tracking
View operations dashboard	Live data aggregation and display
Export reports & log system actions	Report generation, download, audit logging

2.1 User & Access Management

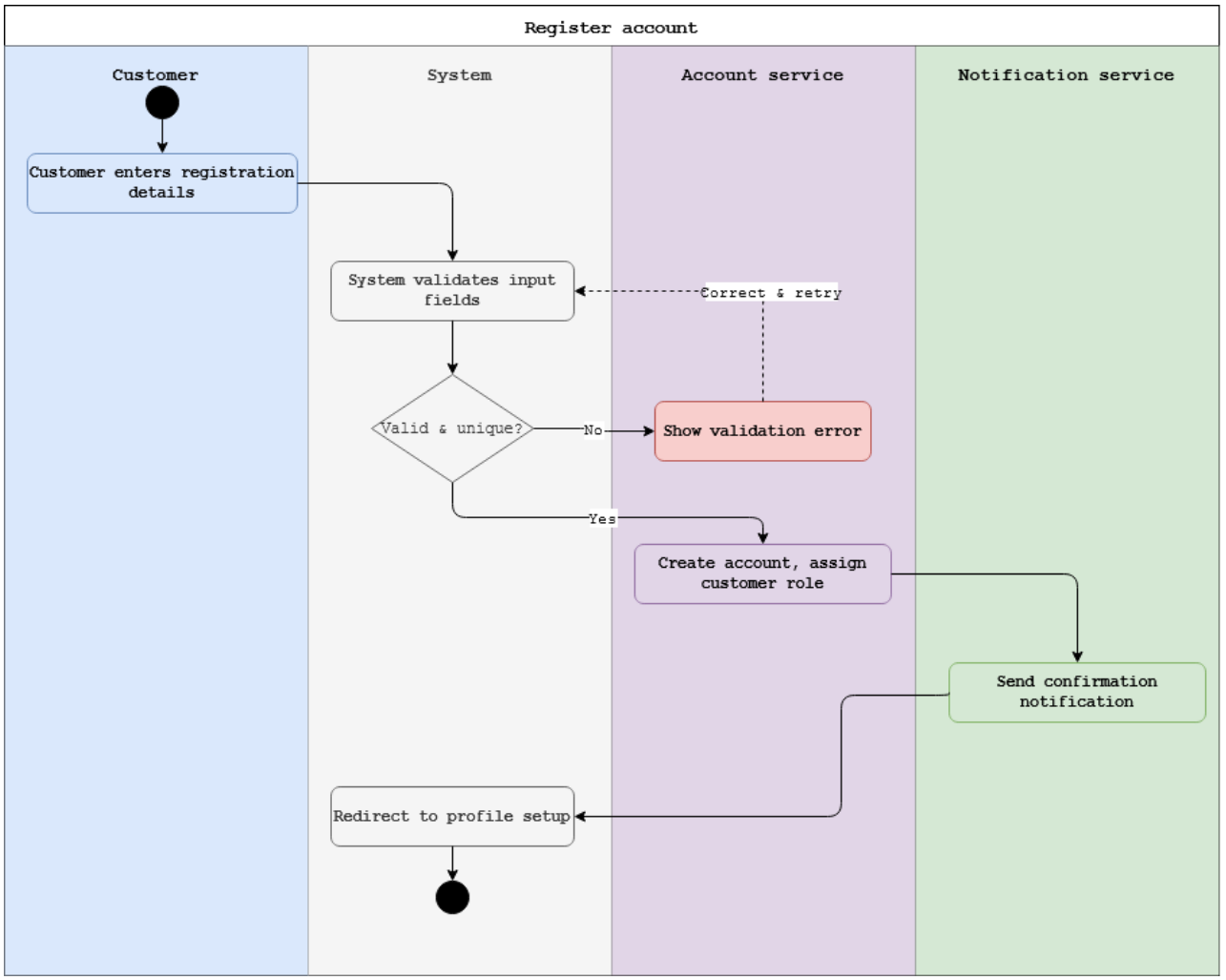
Log in

The user enters credentials. The system validates them and increments a failure counter on each incorrect attempt. After three consecutive failures the account is locked and the user notified. On success the system determines the role and routes to the customer dashboard or the staff portal with appropriate permissions.



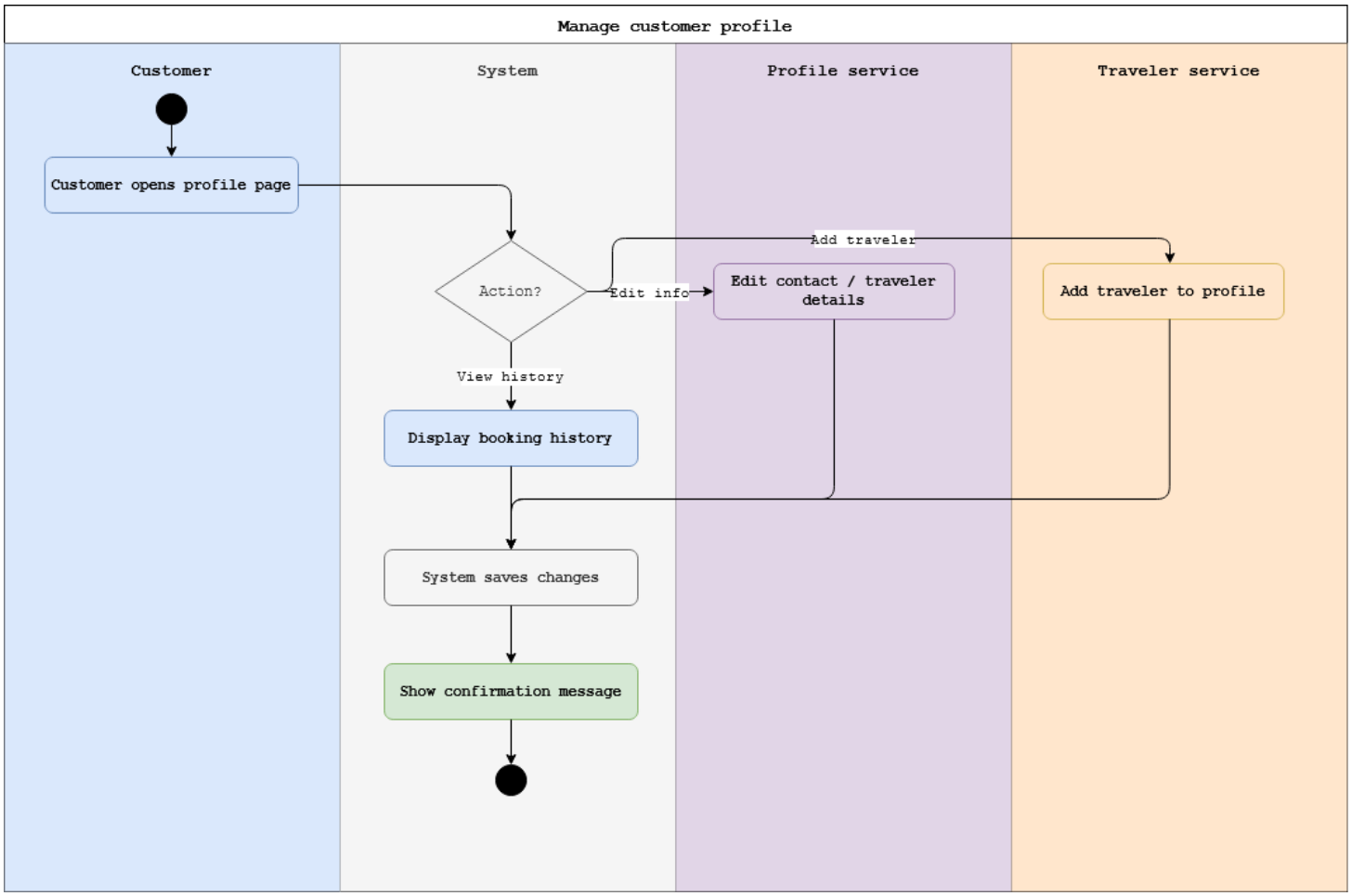
Register account

The customer enters registration details. The system validates all fields for completeness and uniqueness. Errors are surfaced and the customer is prompted to correct them. On success the account is created with a customer role, a confirmation notification sent, and the customer redirected to profile setup.



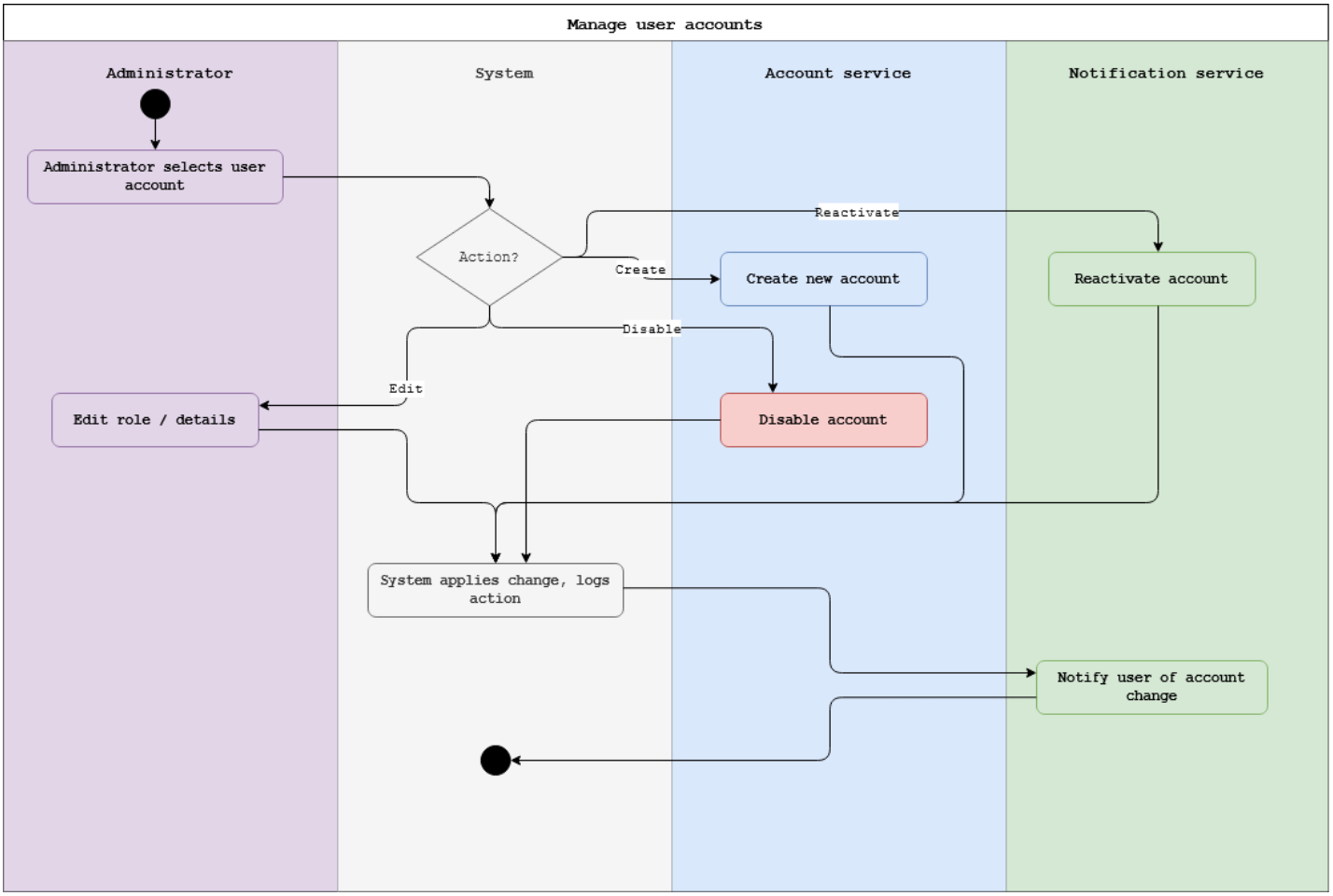
Manage customer profile

The authenticated customer opens their profile and chooses to update contact or traveler details, add an additional traveler, or view booking history. The system saves changes and confirms success.



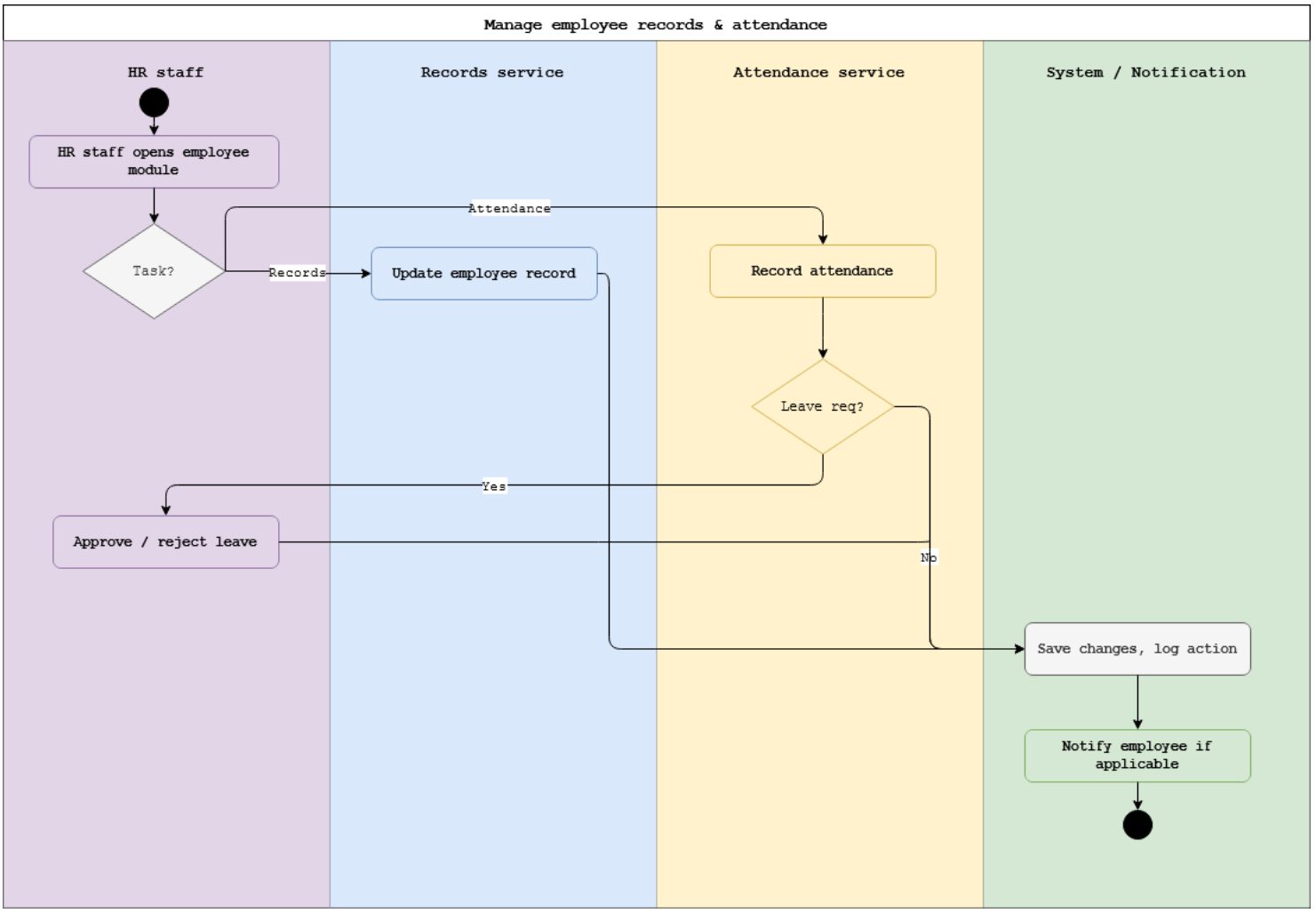
Manage user accounts

The administrator selects a user account and applies one of four actions: create, edit, disable, or reactivate. The system applies the change, logs it to the audit trail, and notifies the affected user.



Manage employee records & attendance

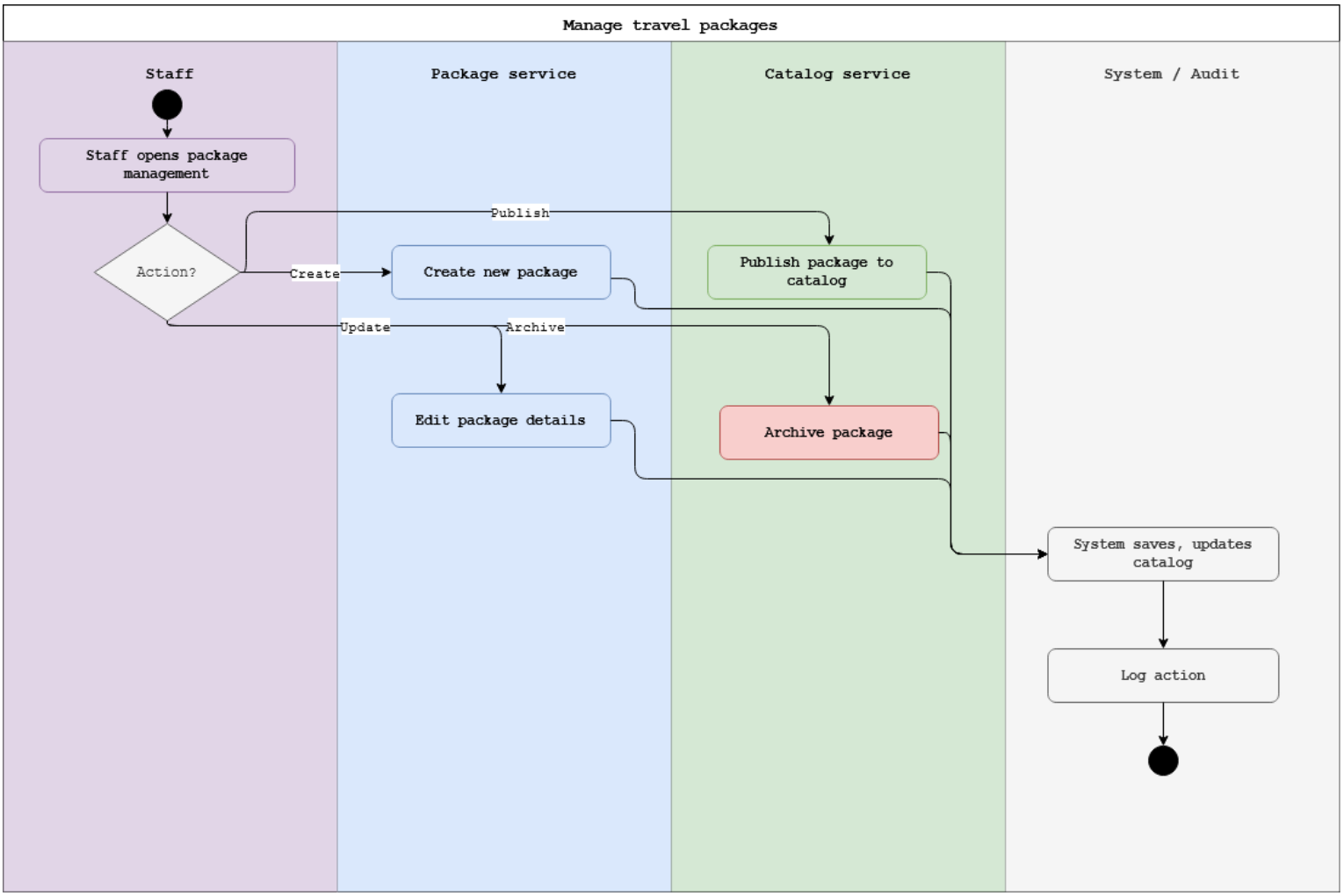
HR staff choose between managing records and managing attendance. For records they update employment details. For attendance they log attendance and, if a leave request is present, approve or reject it. The system saves all changes and notifies the employees where applicable.



2.2 Travel Package Management

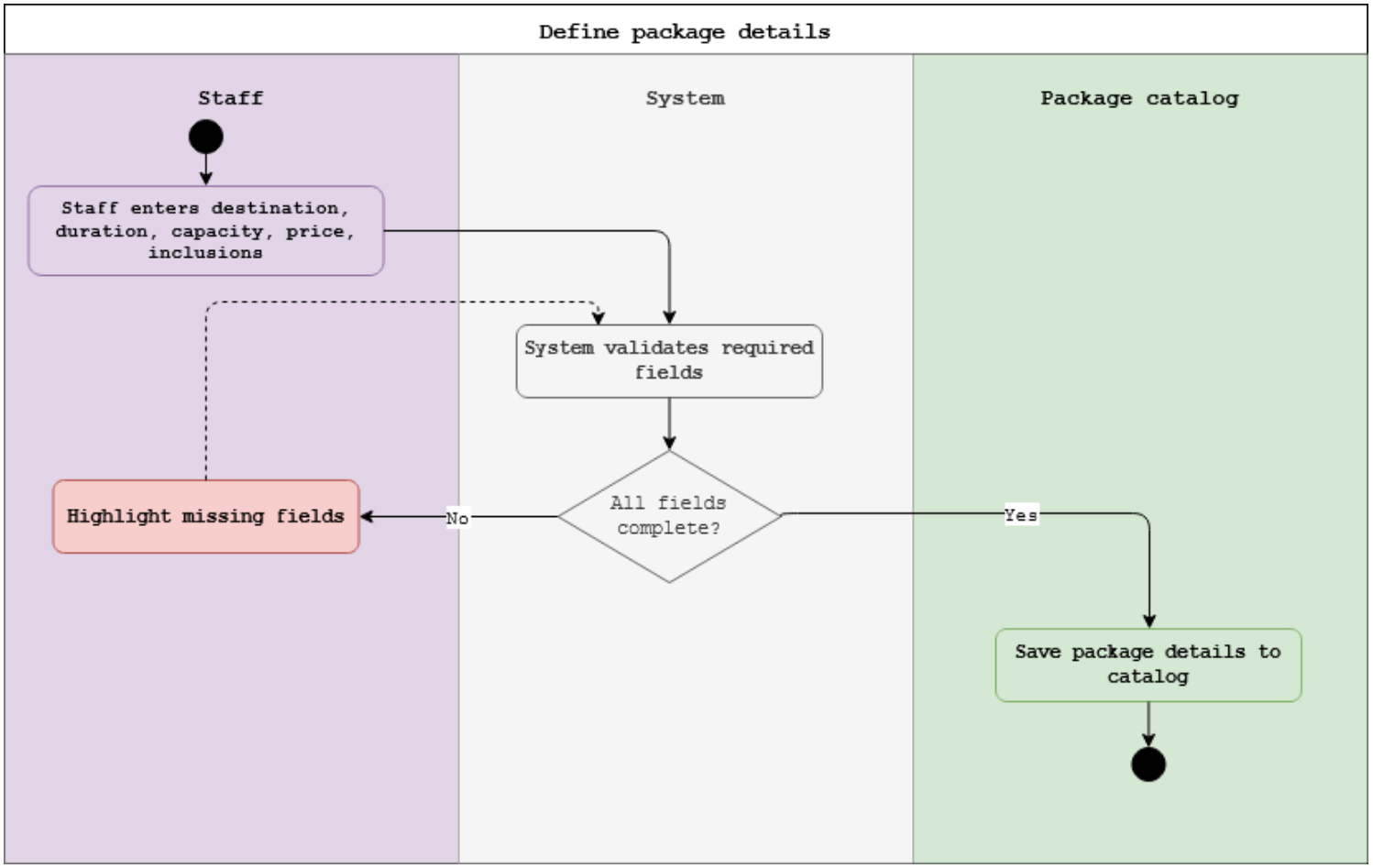
Manage travel packages

Staff choose one of four actions: create a new package, update an existing one, archive a package, or publish it to the catalog. The system saves the change, updates the catalog, and logs the action.



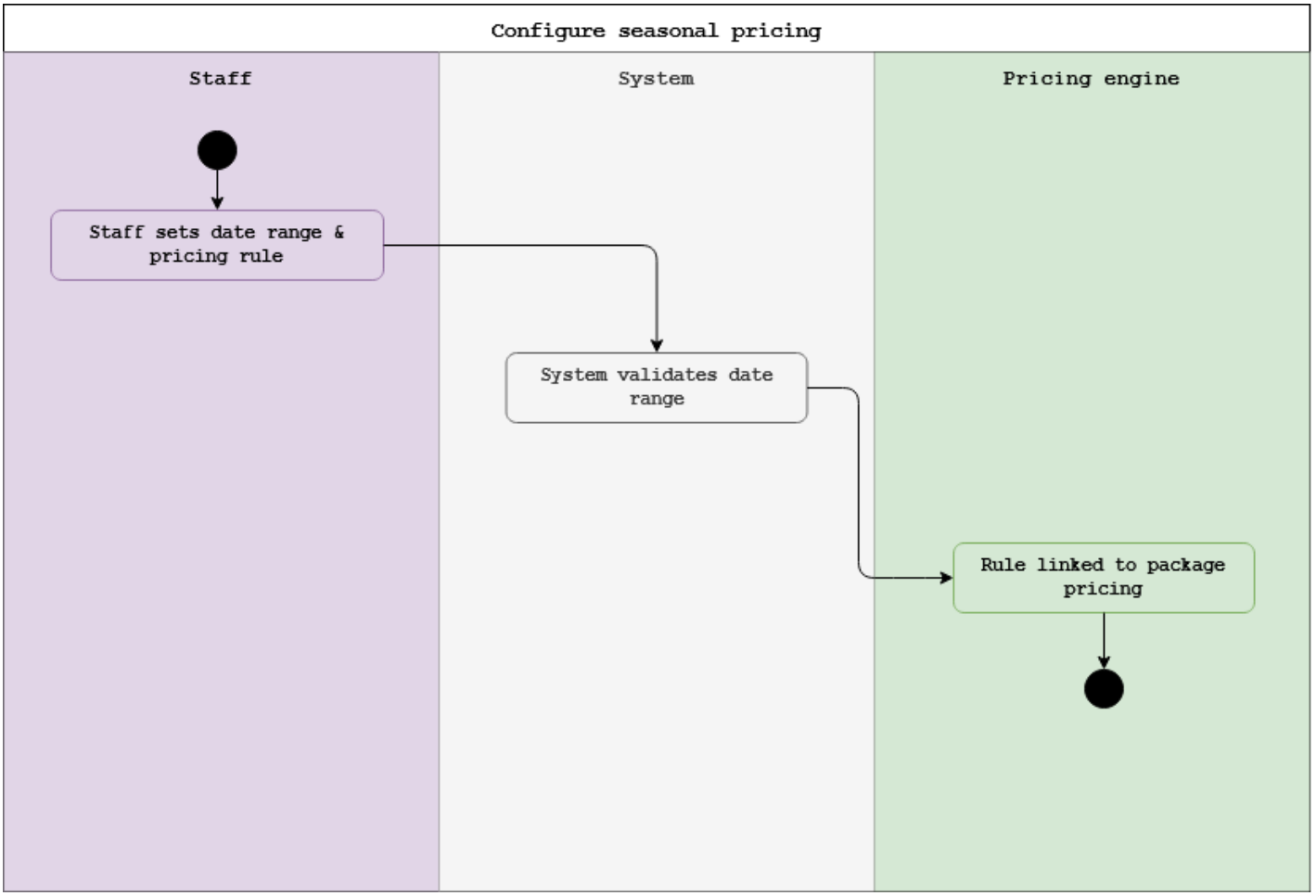
Define package details

Staff enter destination, duration, capacity, price, itinerary, and inclusions. The system validates all required fields are present. Missing fields are highlighted and the staff member prompted to complete them before saving.



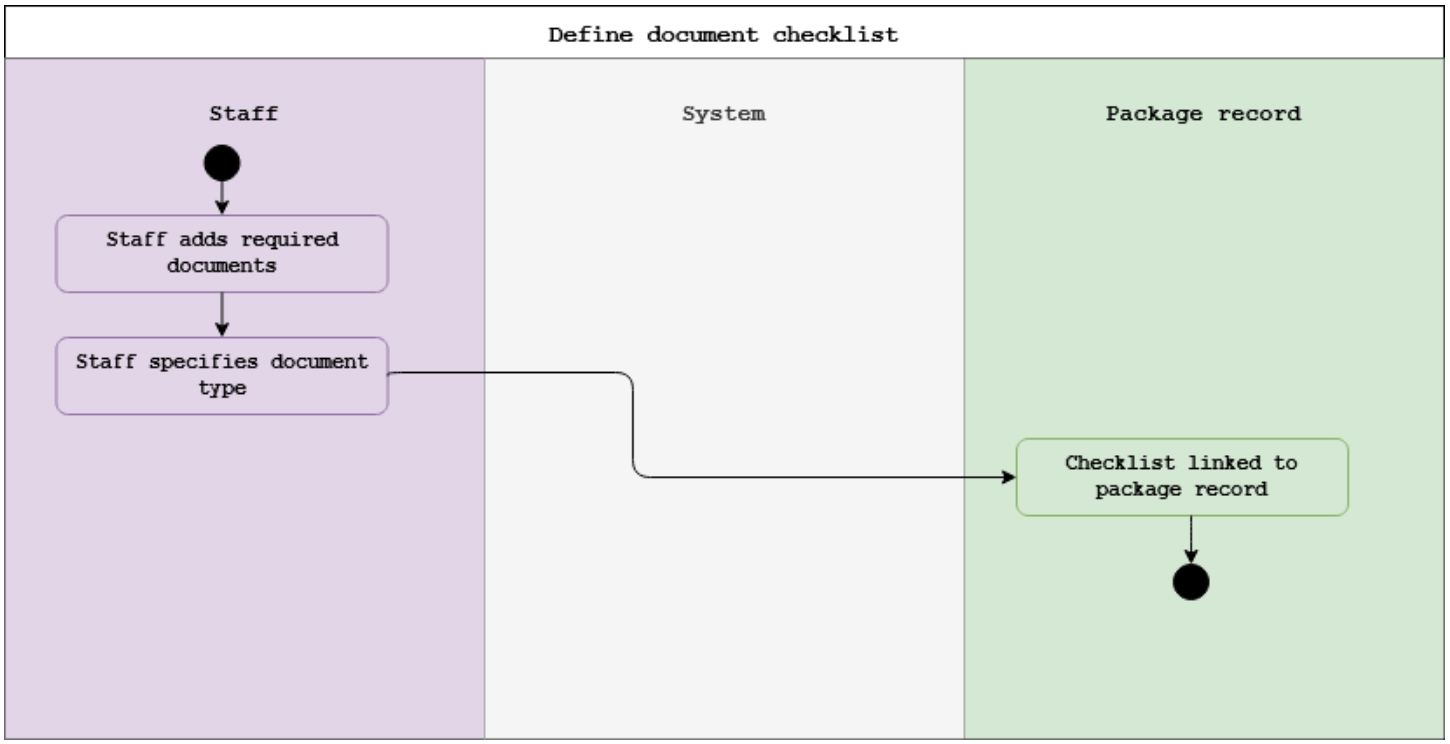
Configure seasonal pricing

Staff define a date range and pricing rule for a package. The system validates the range and links the rule to the package pricing engine.



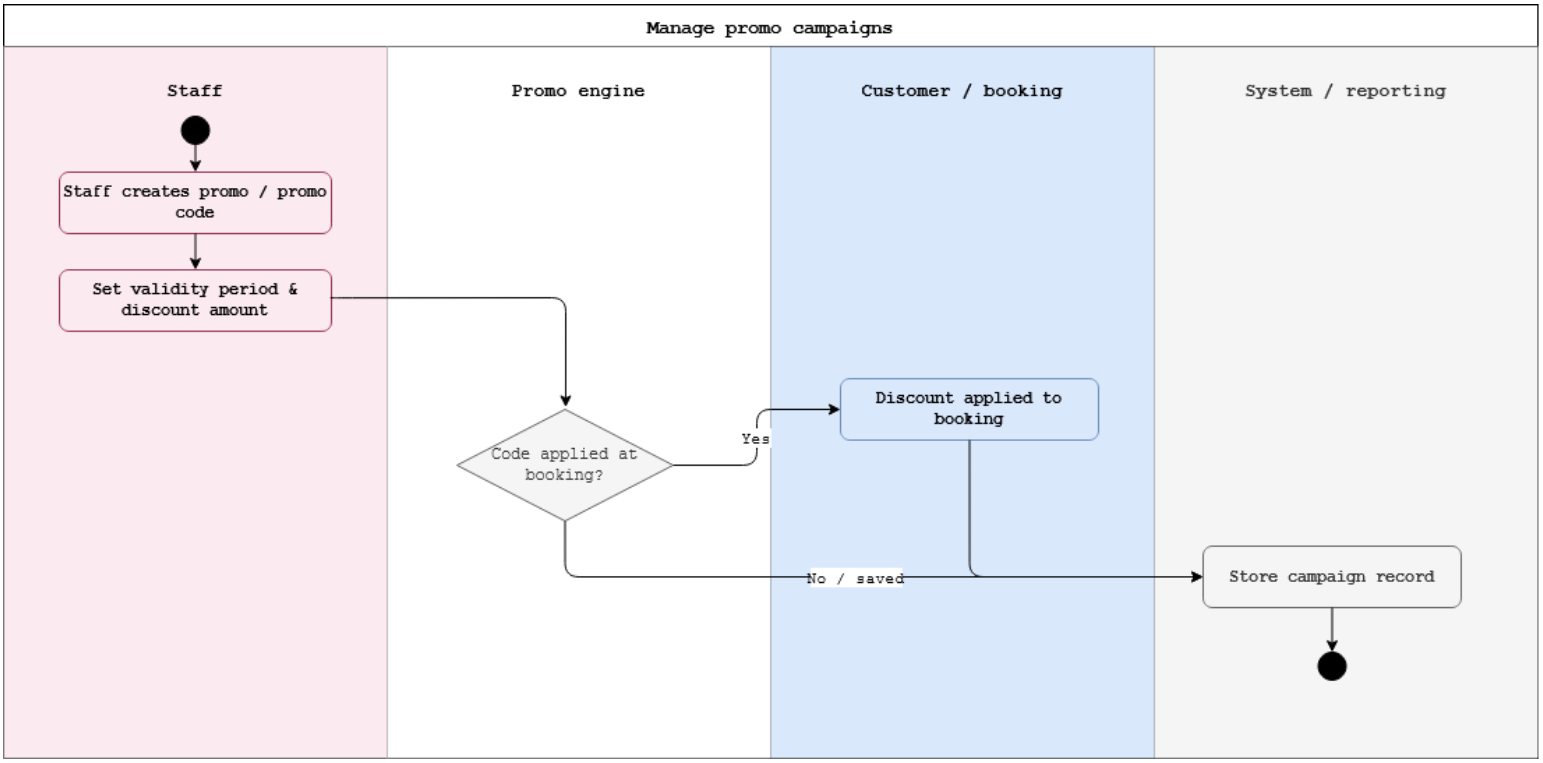
Define document checklist

Staff add the required travel documents for a package. The system links the checklist to the package record for use during booking and trip preparation.



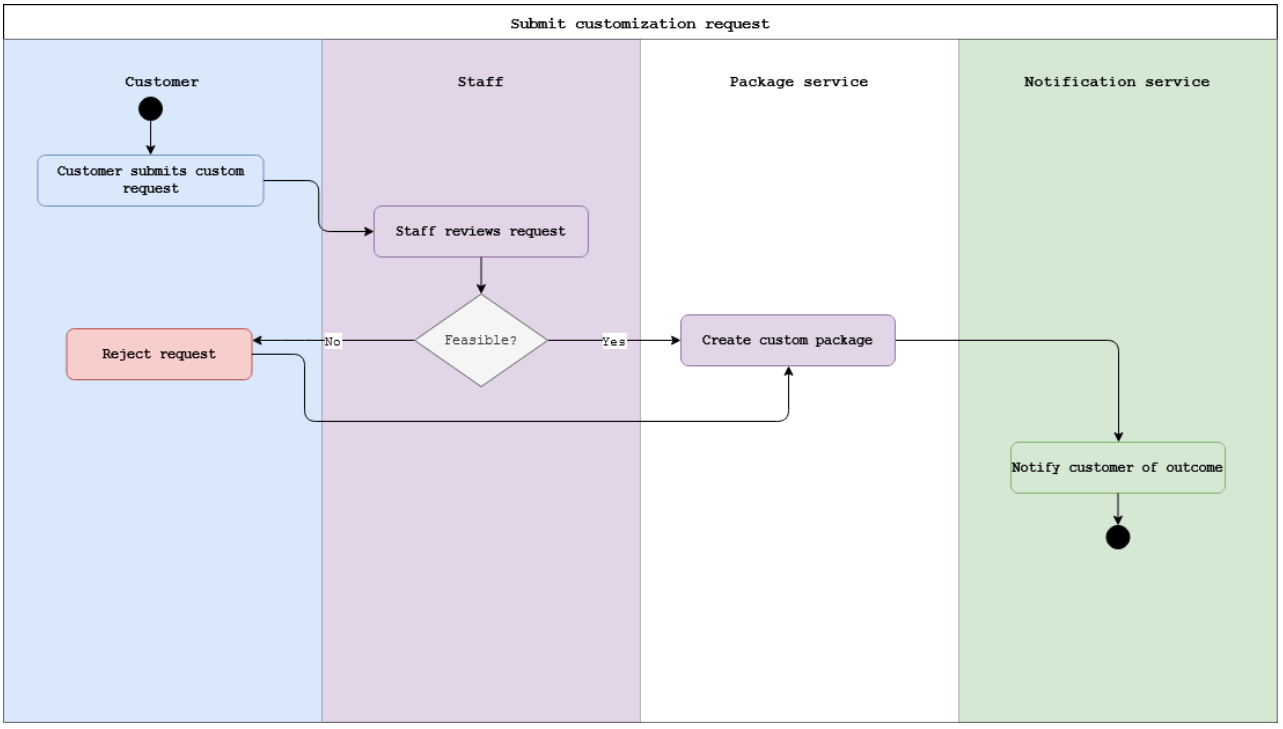
Manage promo campaigns

Staff create a promotional campaign or promo code, setting the validity period and discount amount. When a customer applies the code at booking, the system validates and applies the discount. All campaigns are stored for reporting regardless of use.



Submit customization request

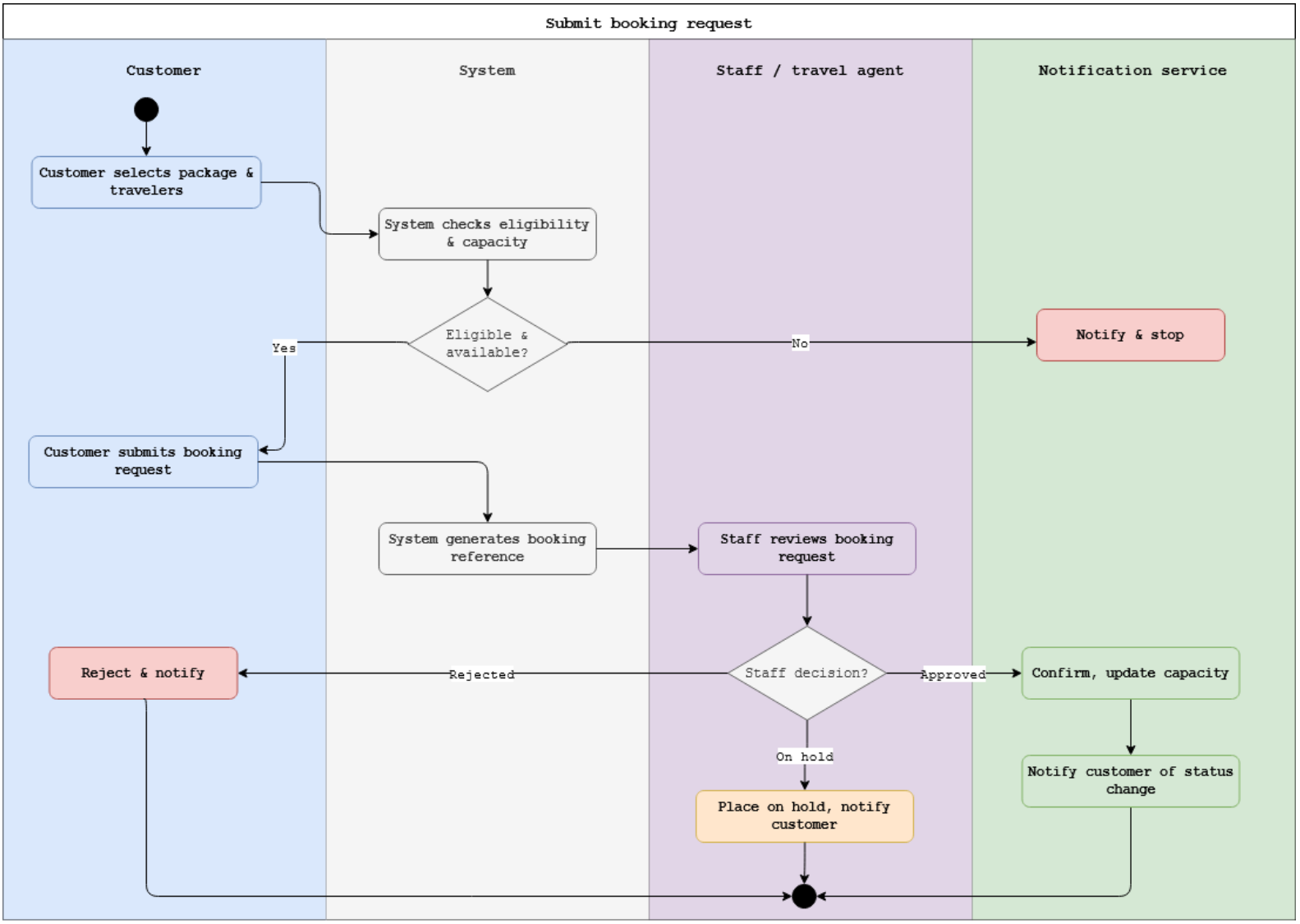
A customer submits a request to customize a package. Staff review the request and assess feasibility. If feasible a custom package is created; if not the request is rejected. The customer is notified of the outcome in both cases.



2.3 Booking & Reservation Management

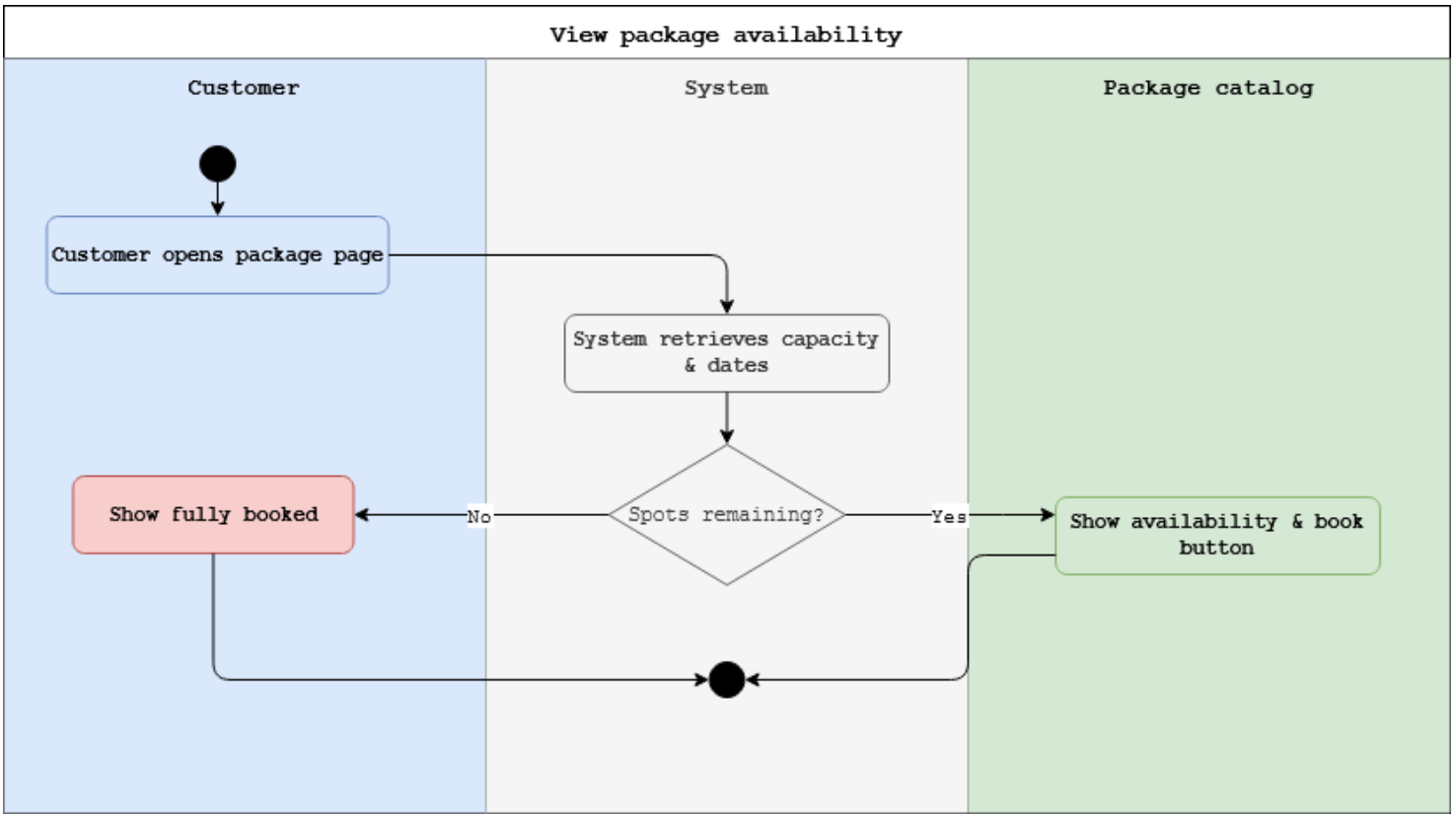
Submit booking request

The customer selects a package and adds travelers. The system checks eligibility and capacity. If either fails, the customer is notified and the flow ends. Otherwise, the request is submitted, a booking reference generated, and staff review it. The outcome is one of three: approved (booking confirmed, capacity updated), rejected (customer notified), or on hold (customer notified). A status notification is sent whenever the booking state changes.



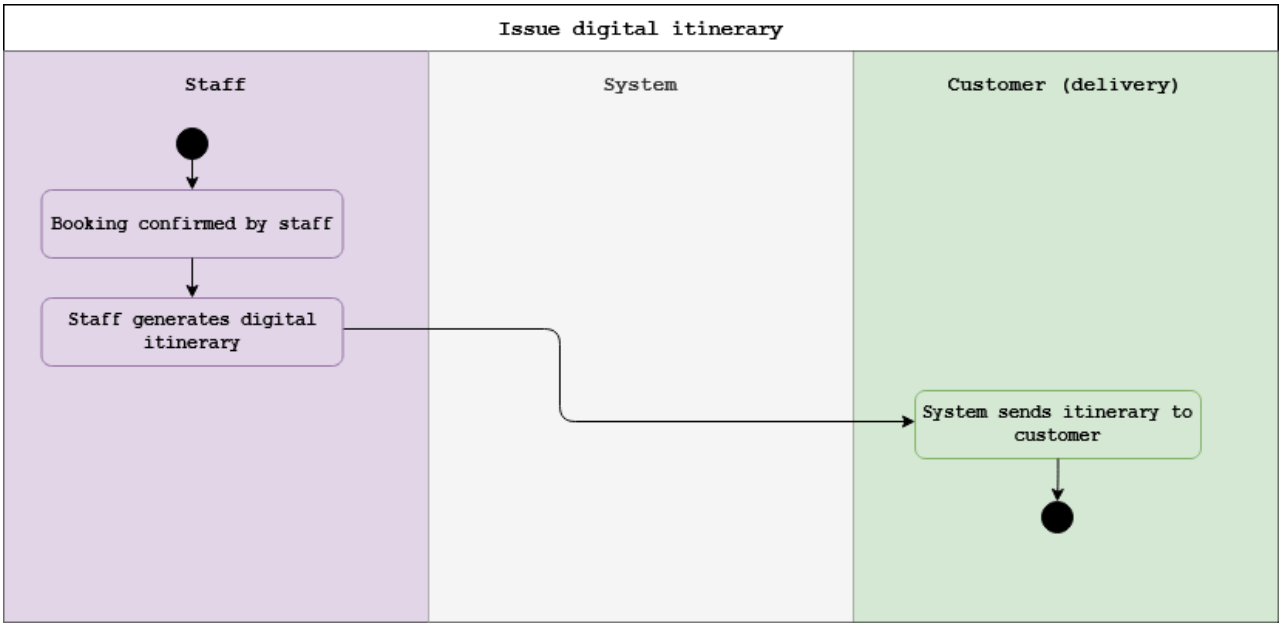
View package availability

The customer opens a package page. The system retrieves current capacity and displays departure dates. If spots remain a booking button is shown; if the package is full a fully booked indicator is displayed instead.



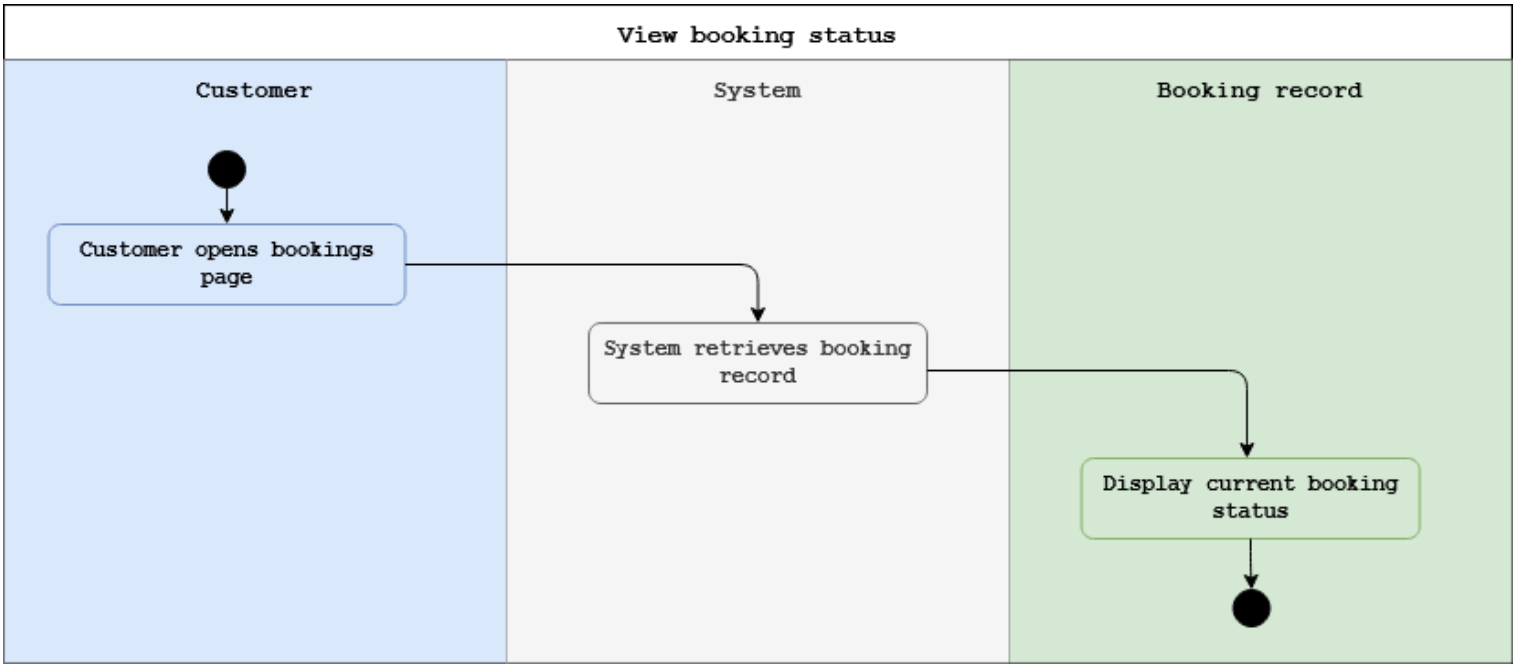
Issue digital itinerary

Once a booking is confirmed, staff generate the digital itinerary for that booking. The system sends it directly to the customer.



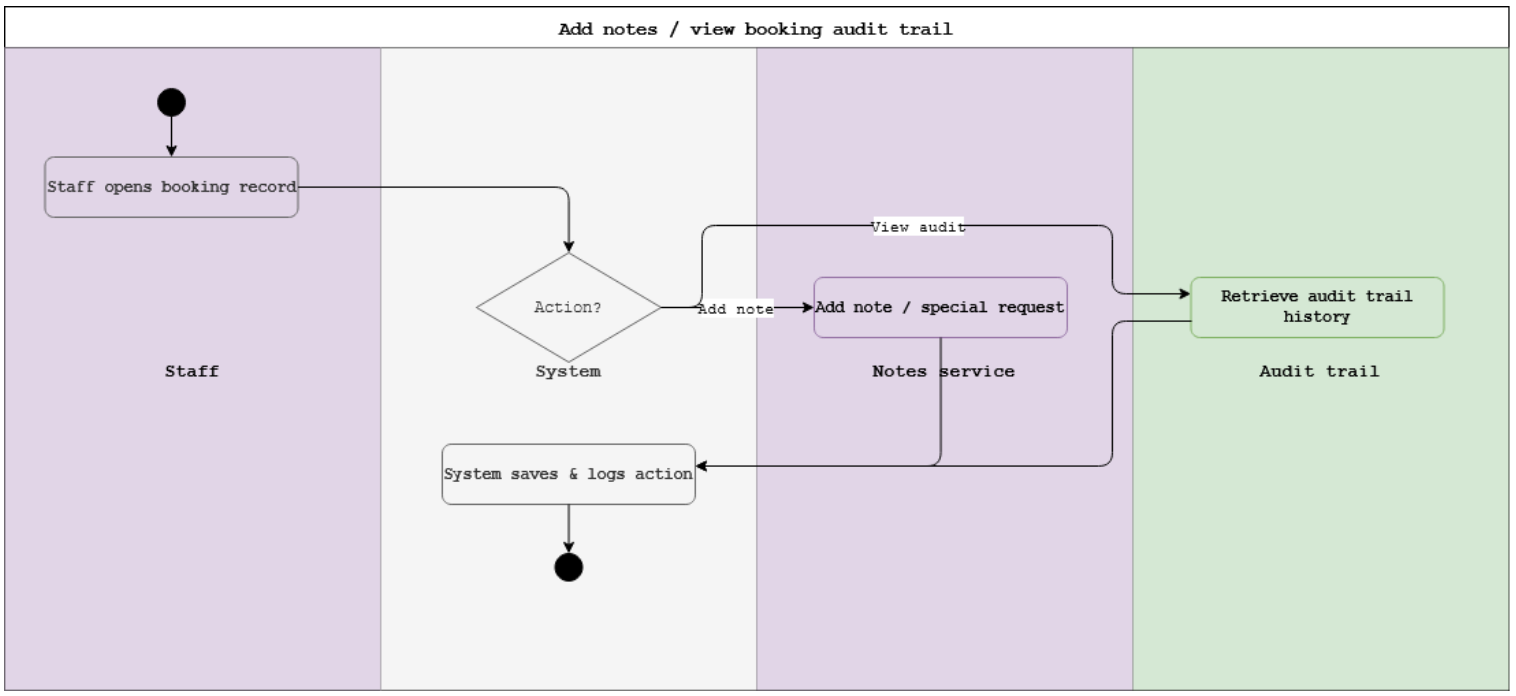
View booking status

The authenticated customers open their bookings page. The system retrieves and displays the current status of the booking, enabling self-service tracking without contacting staff.



Add notes / view booking audit trail

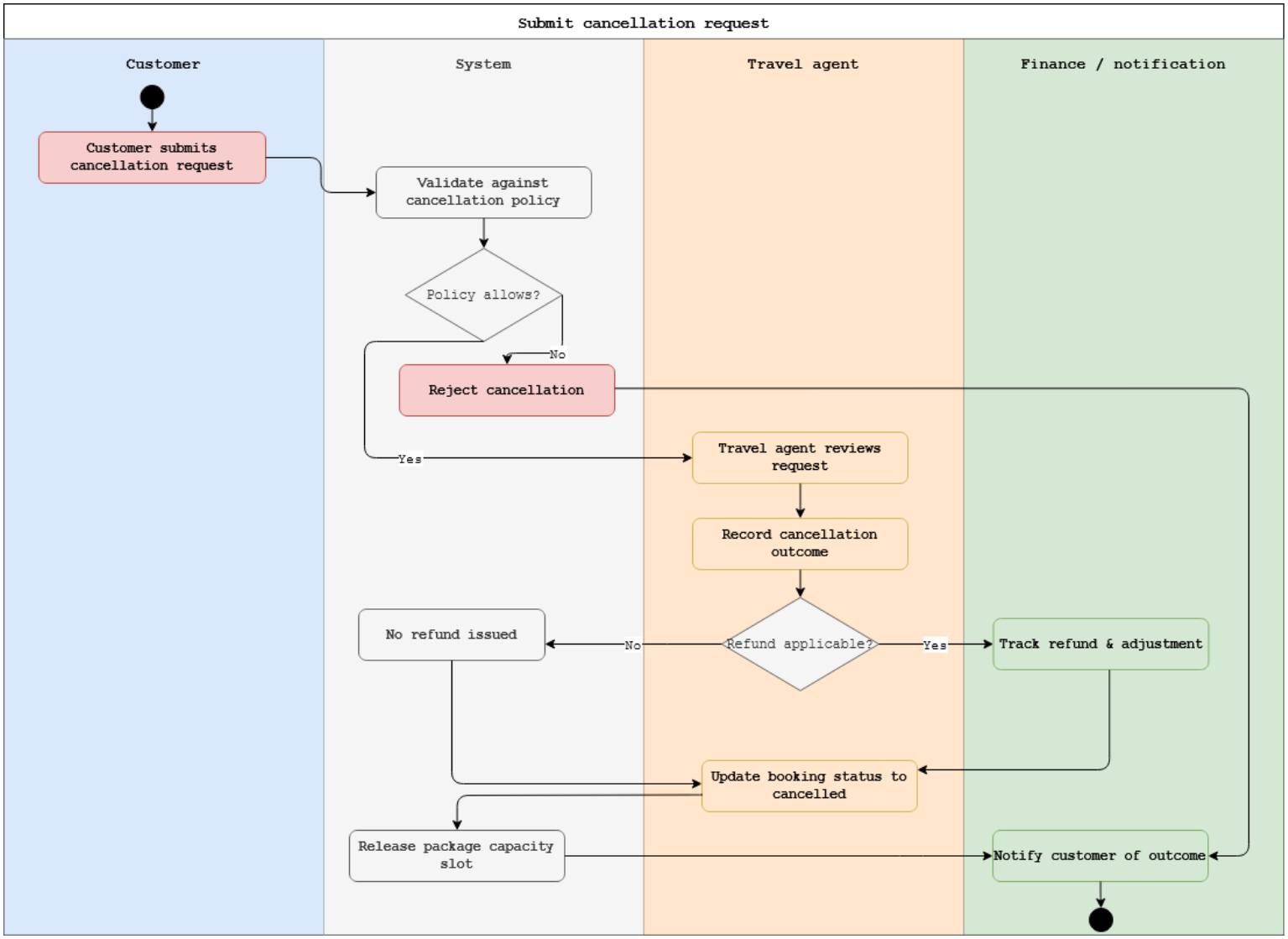
Staff open a booking record and choose to add a note or view the audit trail of major actions taken on that booking. Both actions are saved and contribute to the booking history.



2.4 Cancellation Handling

Submit cancellation request

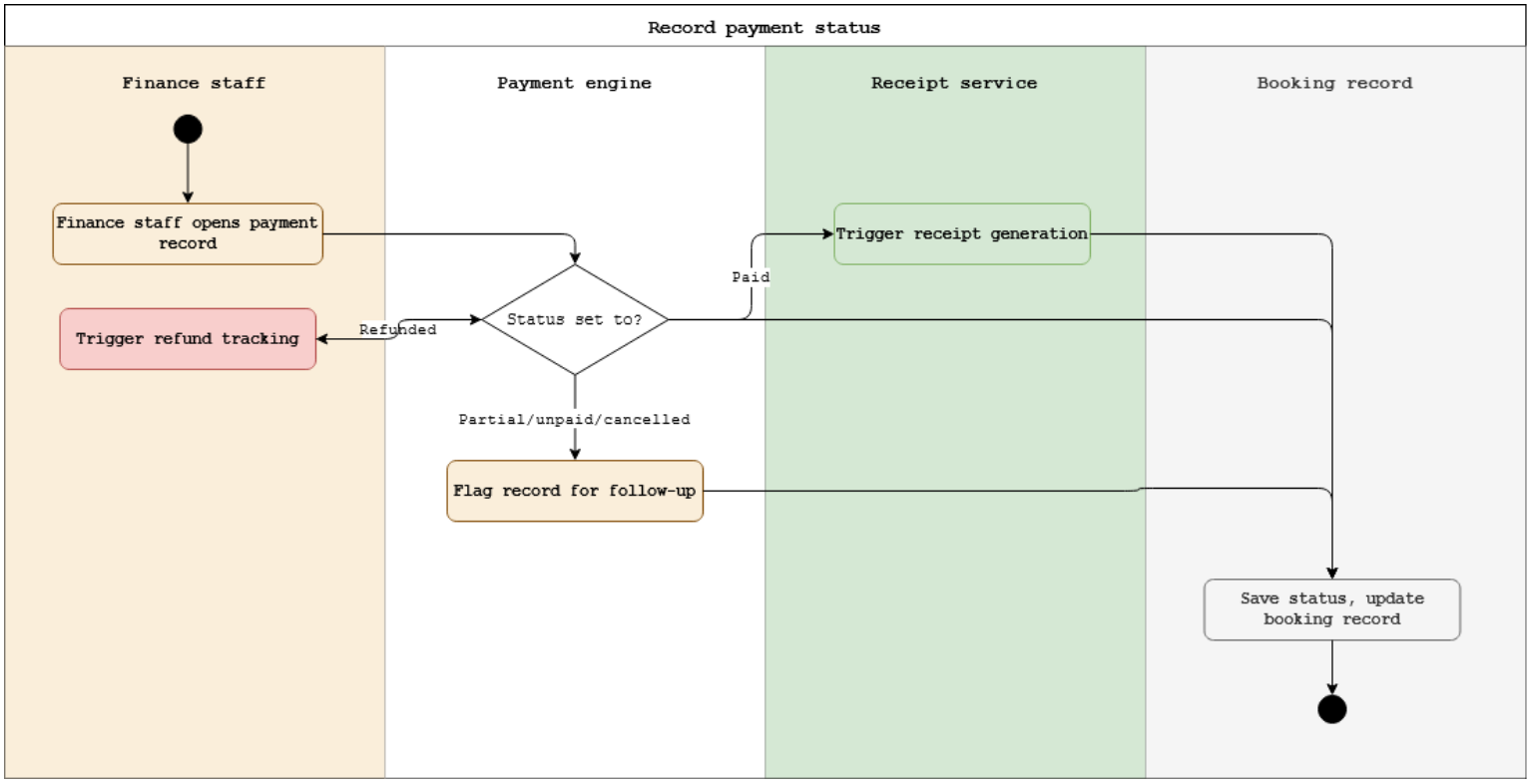
The customer submits a cancellation. The system validates it against the policy. If disallowed the request is rejected and the customer is notified. If allowed a travel agent reviews it, records the outcome, and determines whether a refund applies. The booking status is updated, the capacity slot releases, and the customer is notified of the final outcome.



2.5 Financial Management

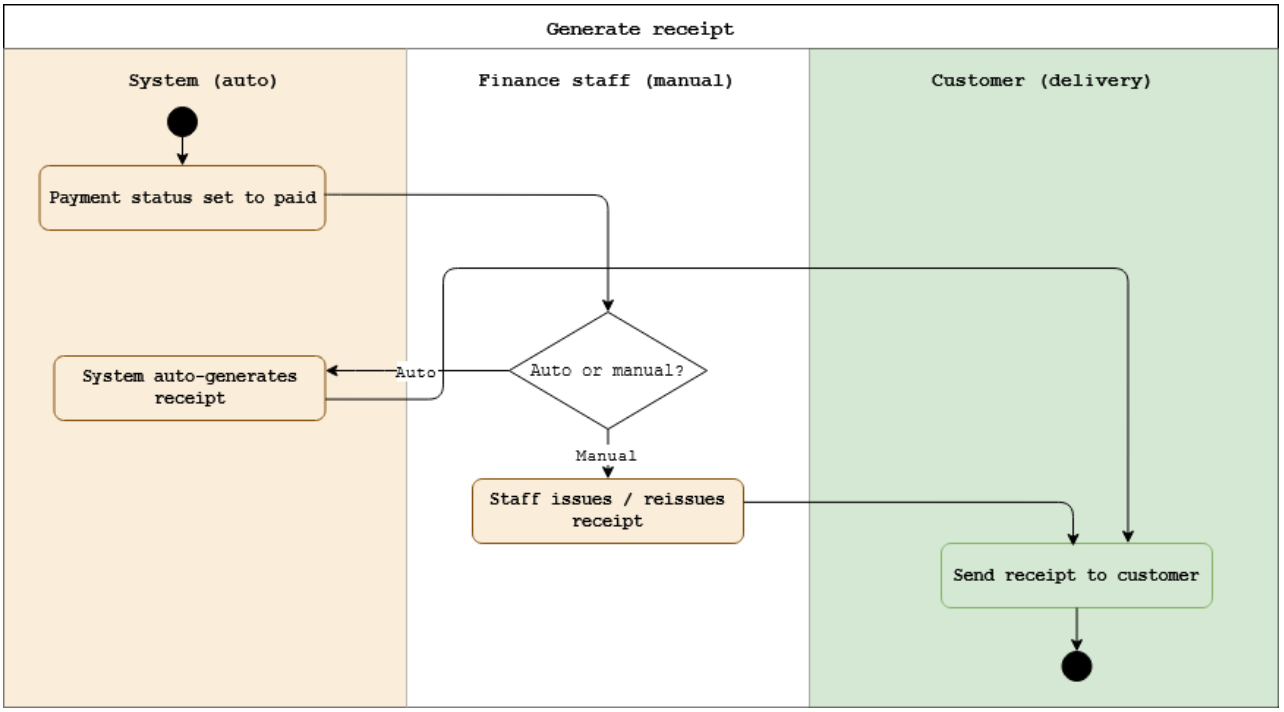
Record payment status

Finance staff set the payment status as one of five values: paid, partial, unpaid, refunded, or cancelled. The “paid” path triggers receipt generation. The “refunded” value triggers refund tracking. “partial”, “unpaid” and “cancelled” flag the record for follow-up.



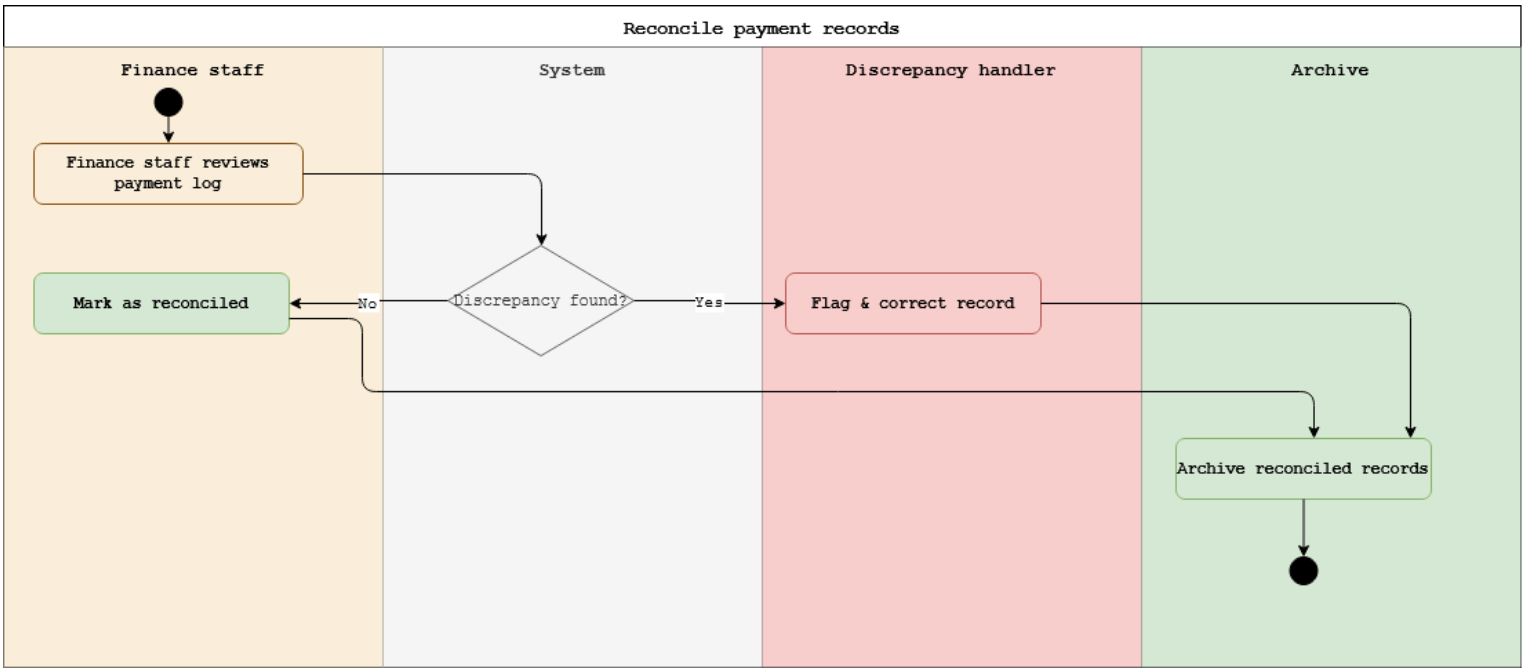
Generate receipt

When a payment is marked as paid the system can generate a receipt automatically, or staff can manually issue or reissue one. In both cases the receipt is sent to the customer.



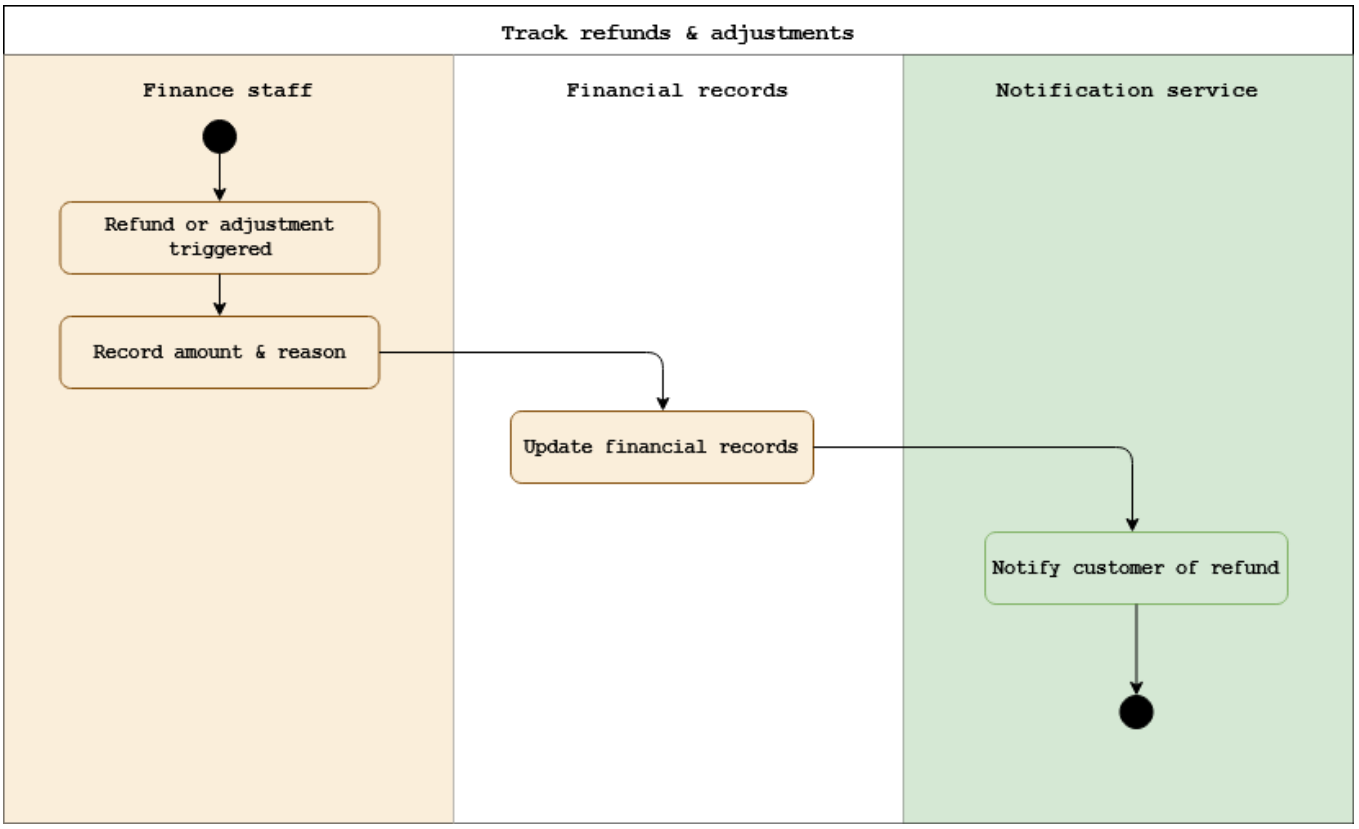
Reconcile payment records

Finance staff review the payment log. If a discrepancy is found the record is flagged and corrected. If no discrepancy exists, the record is marked, reconciled and archived.



Track refunds & adjustments

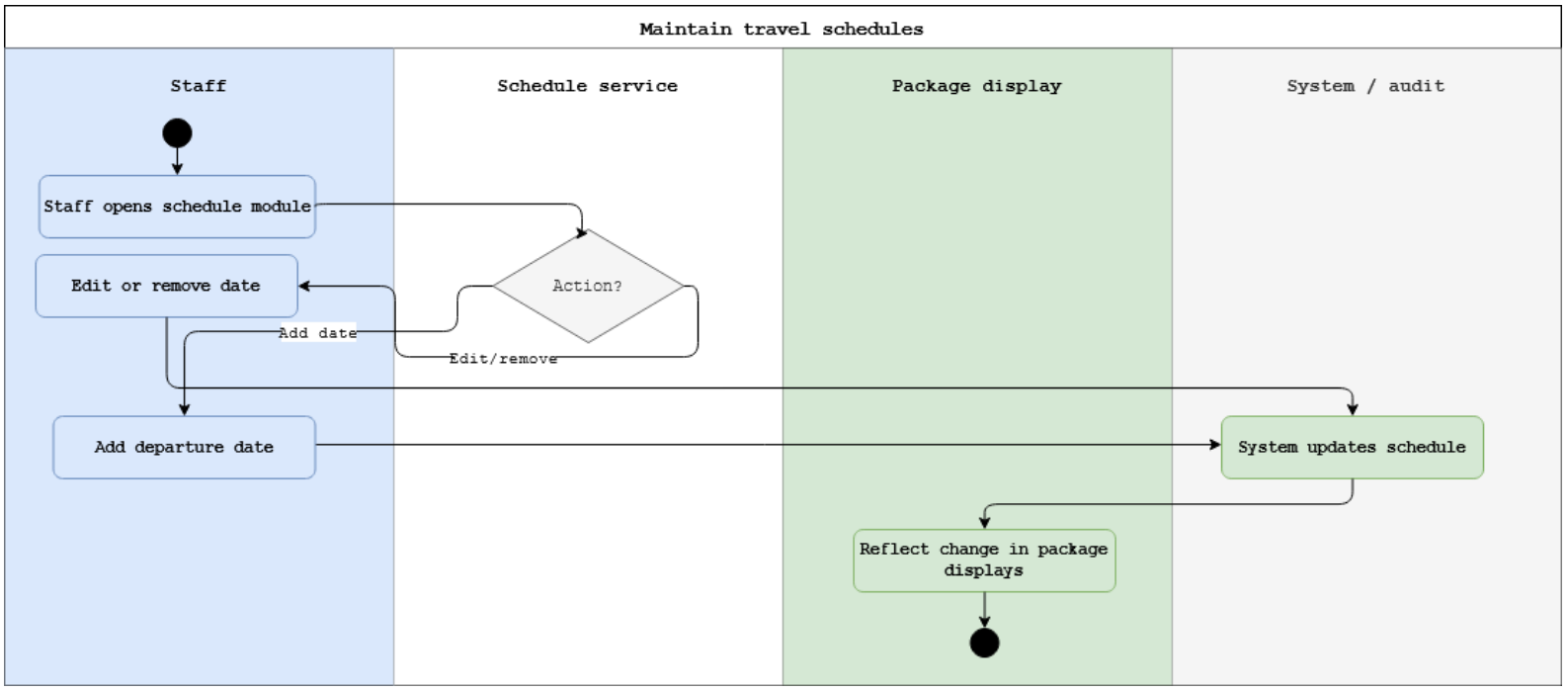
When a refund or adjustment is triggered, finance staff record the amount and reason. The system updates financial records and notifies the customer.



2.6 Scheduling & Operations

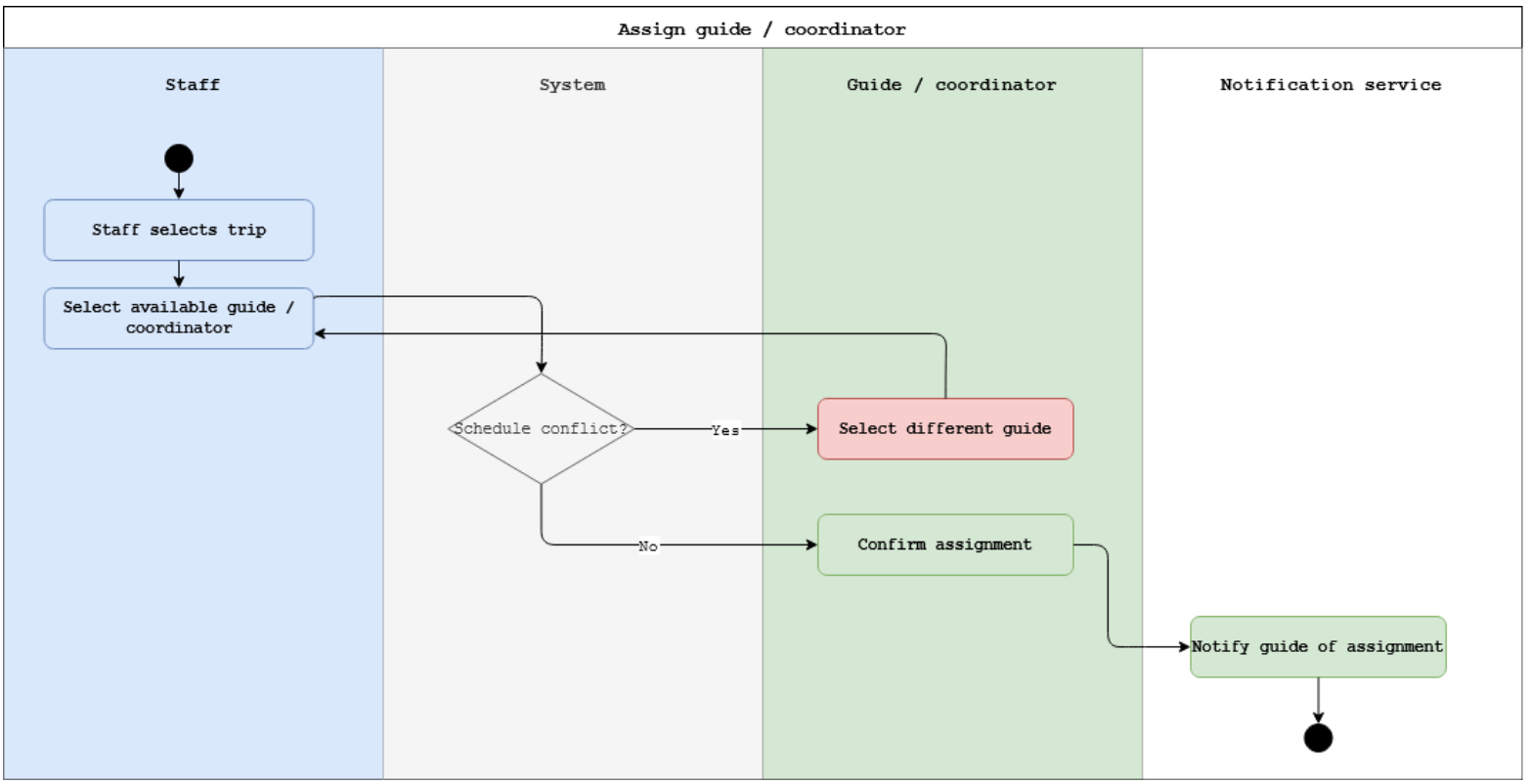
Maintain travel schedules

Staff choose to add a new departure date, edit an existing date, or remove a date. The system updates the schedule and reflects the change across all relevant package displays.



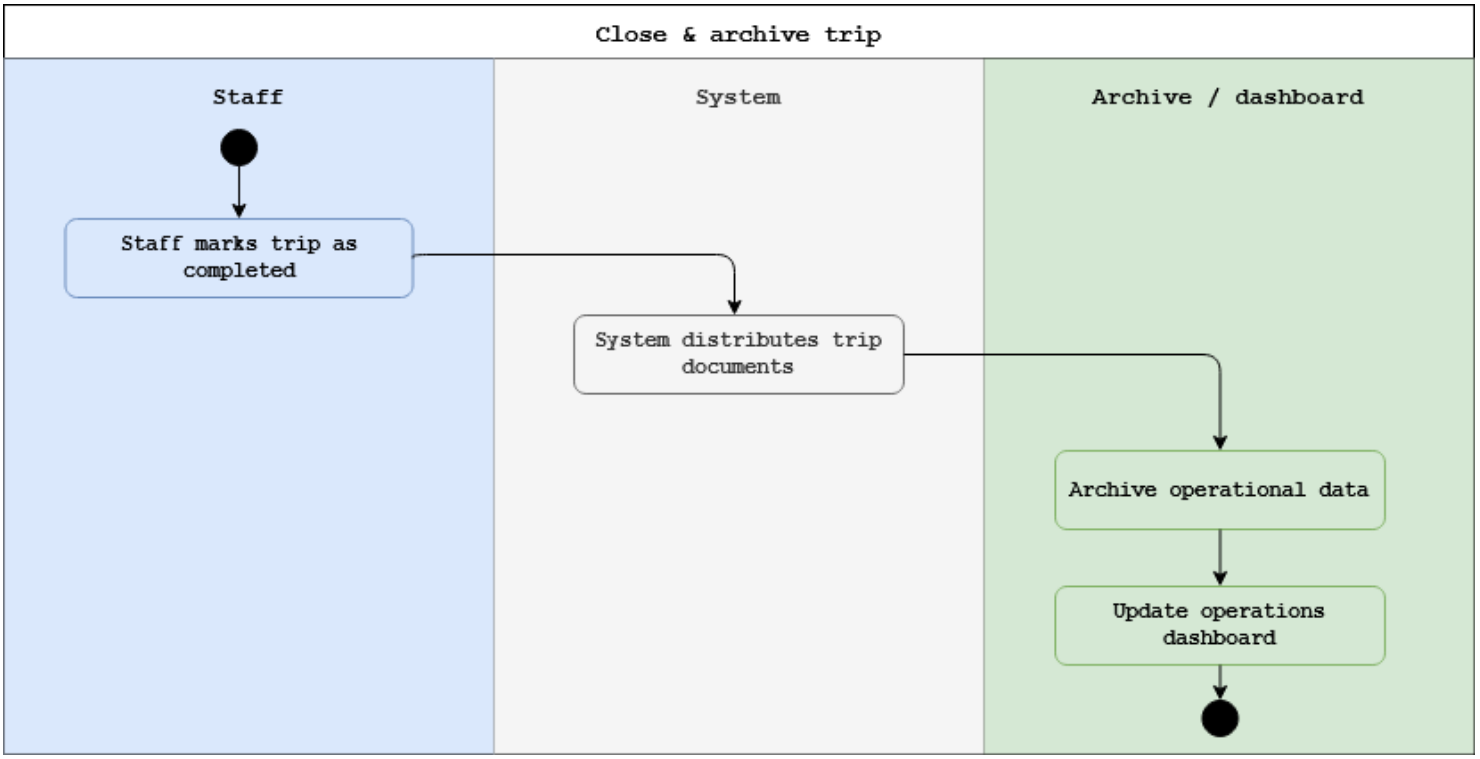
Assign guide / coordinator

Staff select a trip and assign a guide or coordinator. The system checks for scheduling conflicts. If a conflict exists with staff, select a different guide. Once confirmed the assignment is saved and the guide notified.



Close & archive trip

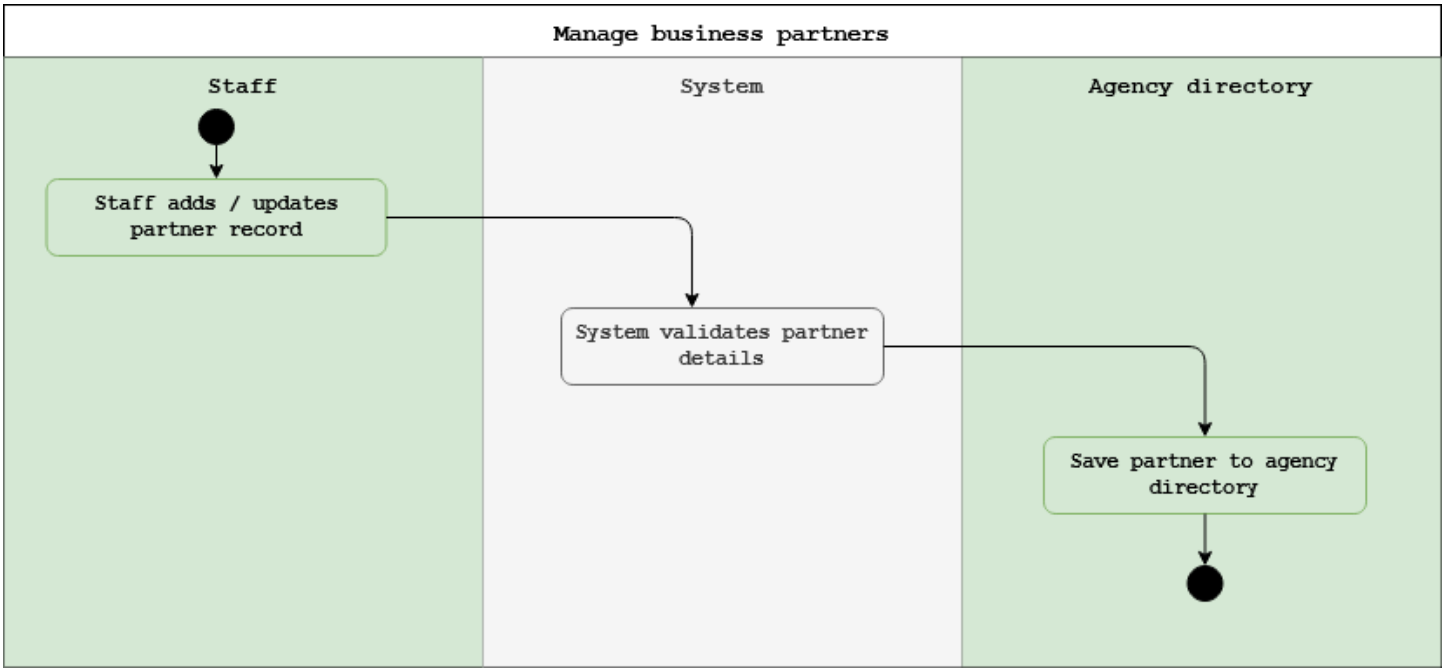
Staff mark a trip as completed. The system automatically distributes required trip documents to internal personnel (include relationship), archives the operational data, and updates the operations dashboard.



2.7 Partner Management

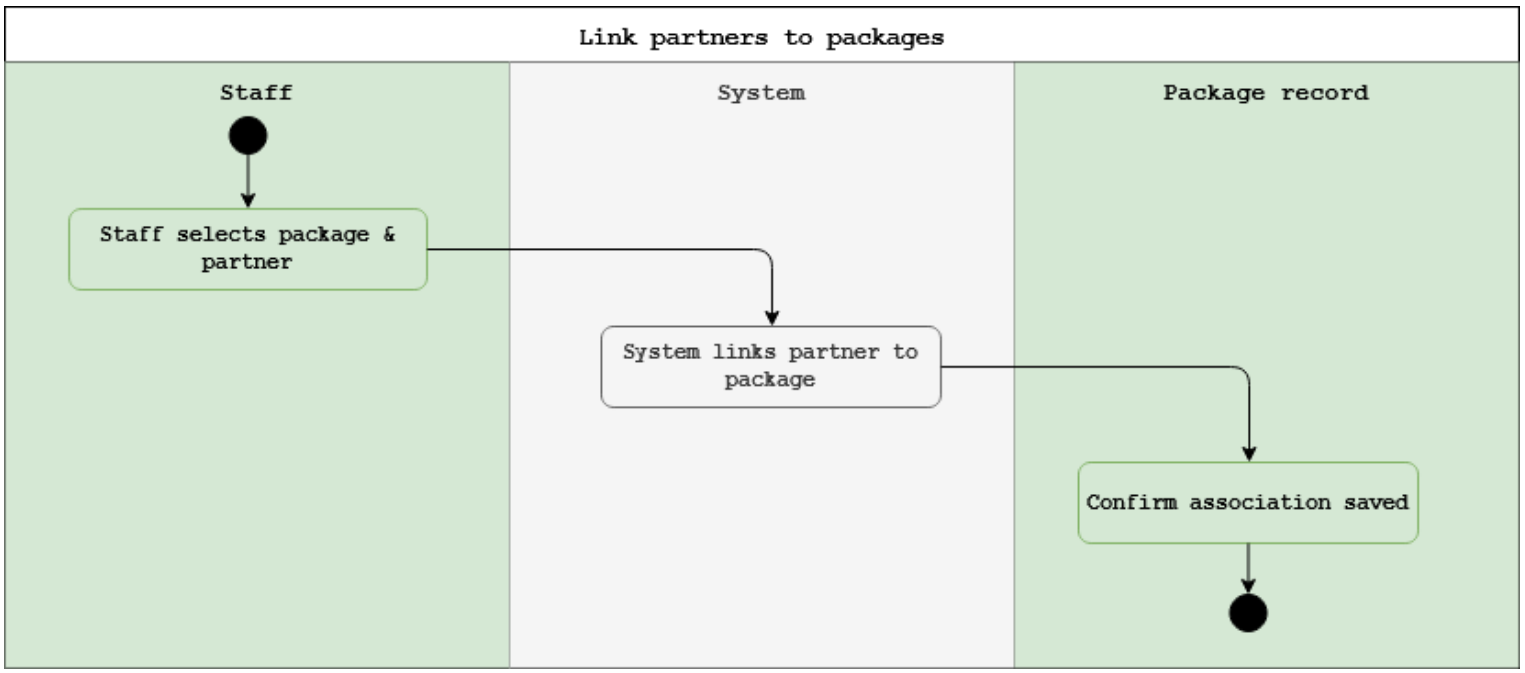
UC-25 Manage business partners

Staff add a new partner or update an existing record. The system validates the details and saves the partner to the agency directory.



UC-26 Link partners to packages

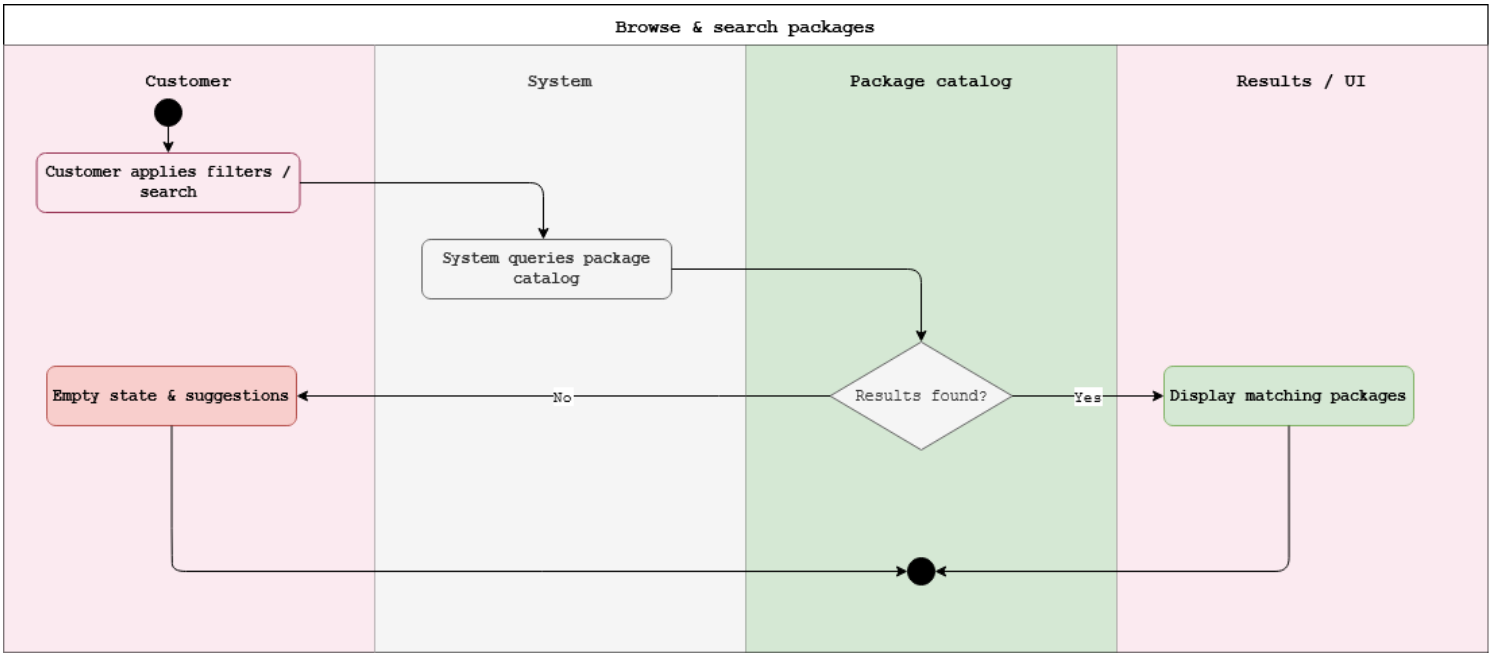
Staff select a package and a partner and link them. The system records the association and confirms the saving.



2.8 Customer Experience

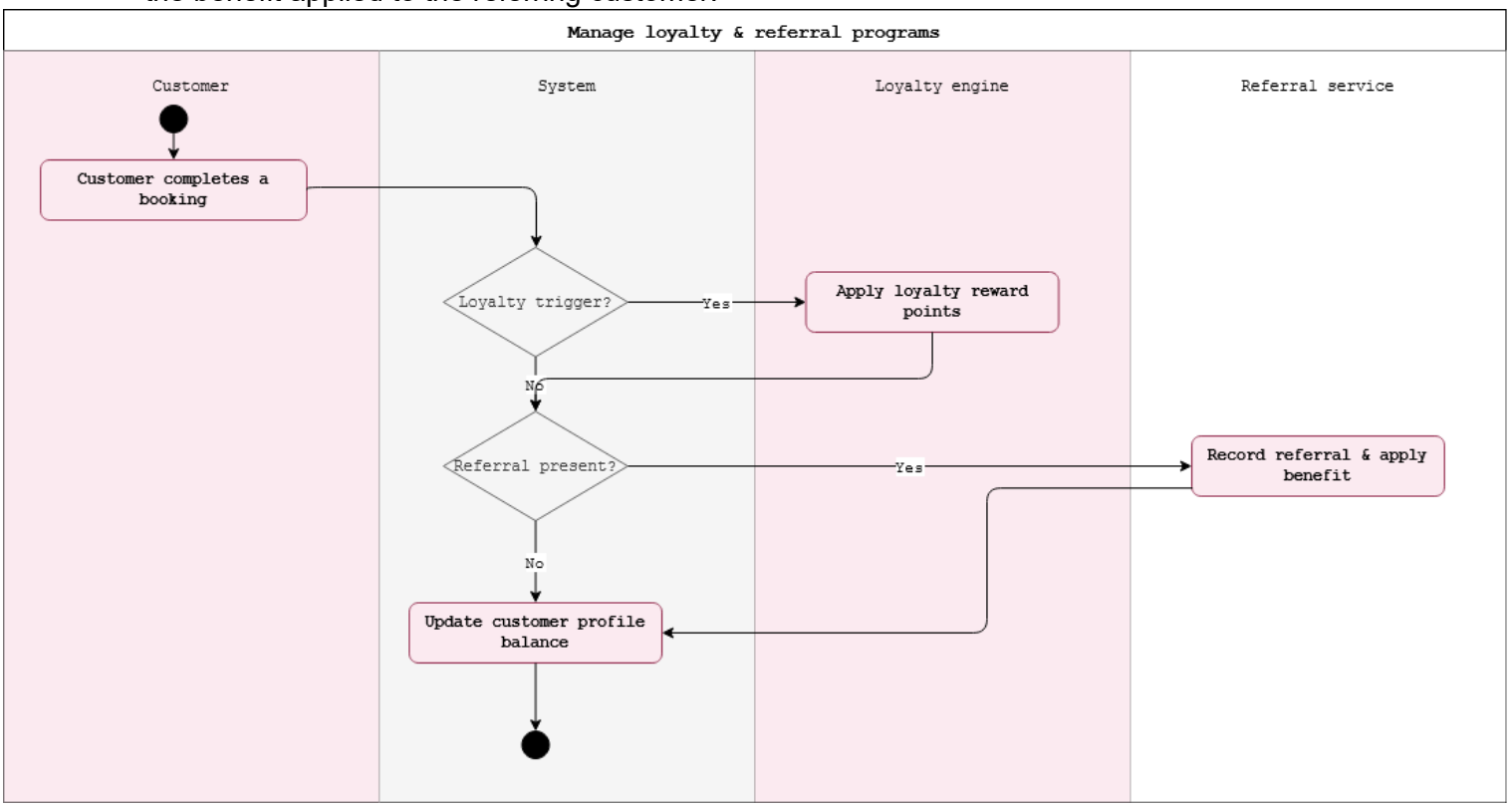
Browse & search packages

The customer applies to search for filters. The system queries the catalog and returns matching published packages. If results are found, they are displayed; if none match an empty state is shown with suggestions.



Manage loyalty & referral programs

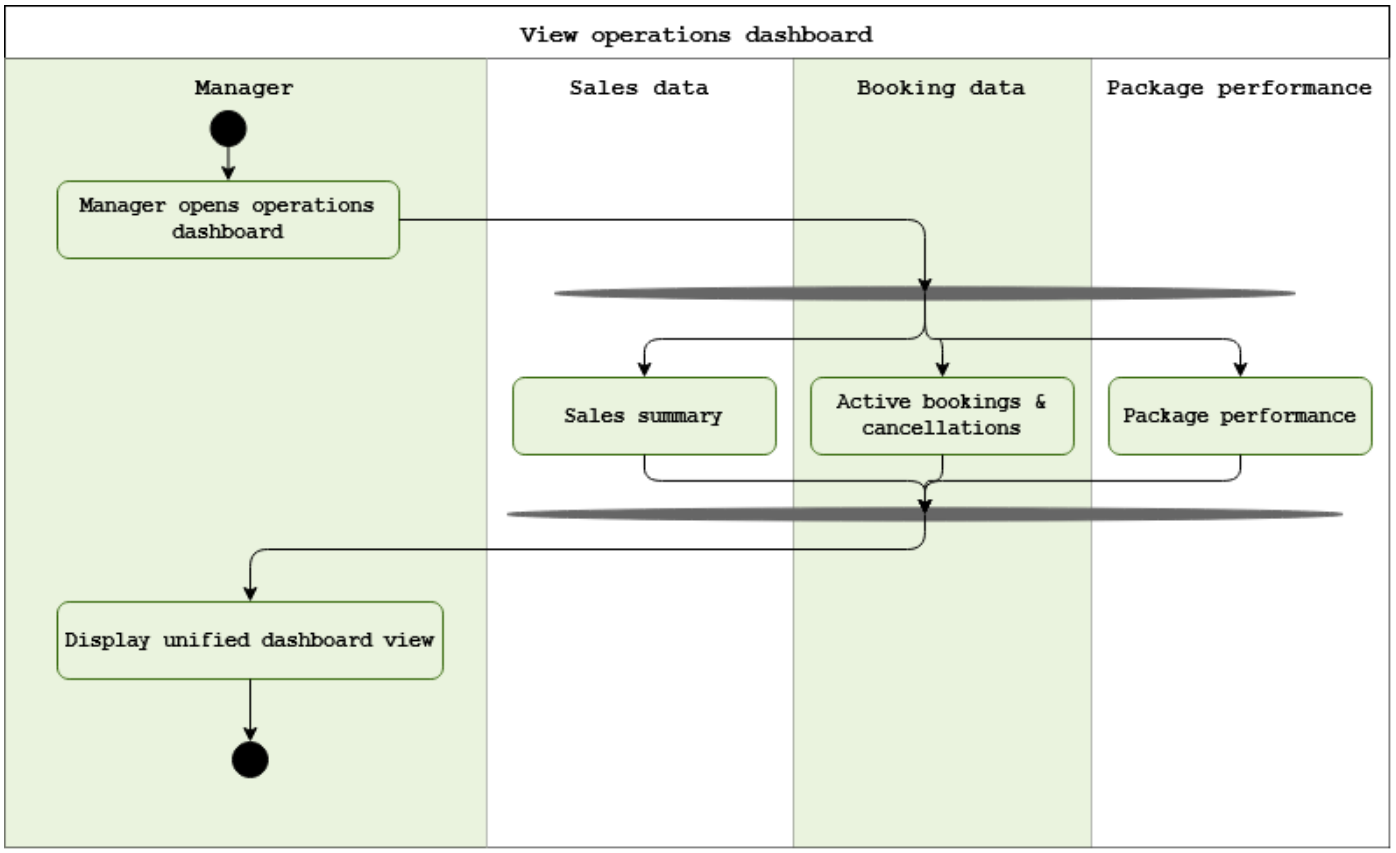
When a booking is completed the system checks for applicable loyalty rewards and, if present, credits them to the customer profile. If the booking was via a referral the referral is recorded and the benefit applied to the referring customer.

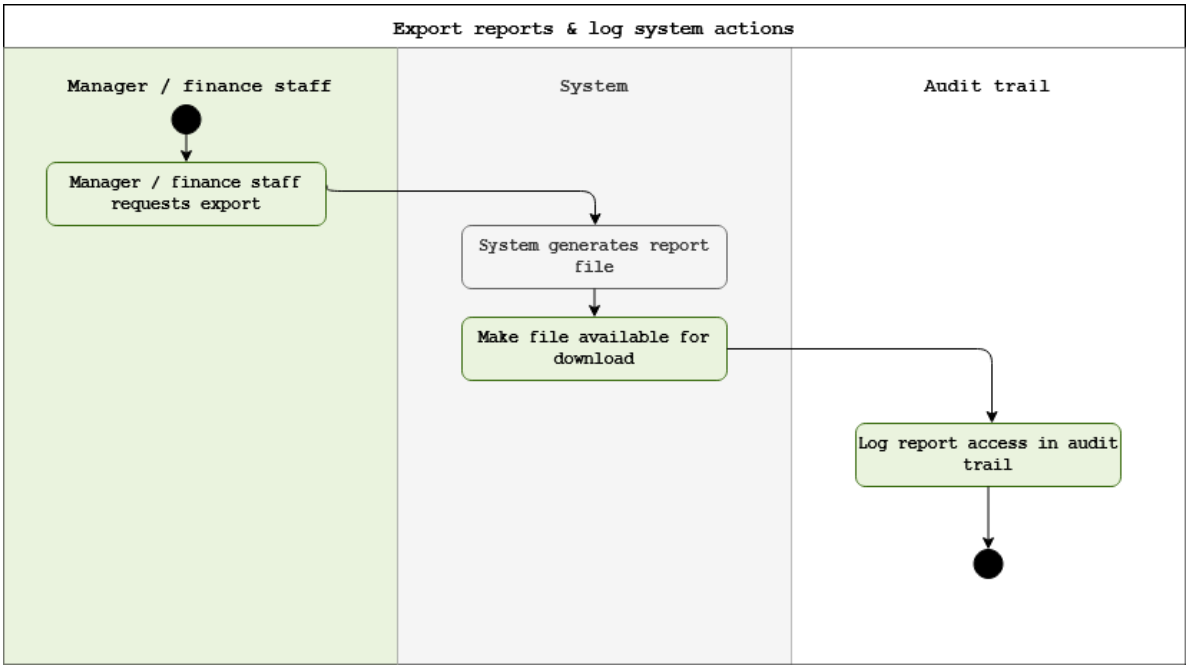


2.9 Reporting & Analytics

View operations dashboard

The manager opens the dashboard. The system aggregates live data in parallel across sales, active bookings and cancellations, and package performance, then displays a unified view.



Export reports & log system actions

The manager or finance staff request an export report. The system generates the file, makes it available for download, and logs the access event to the audit trail.